

# Estimation and Testing for Dependence in Market Microstructure Noise

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## ABSTRACT

This paper proposes new test statistics for the dependence and cross and auto covariance estimators of bivariate noise processes. It derives their asymptotic distributions and provides additional tests for the statistical significance of covariance estimators. Monte Carlo simulation shows that the covariance estimators and test statistics perform better in a finite sample. Further evidence from empirical illustration suggests that the covariance estimators and proposed test statistics are capable of capturing various dependence patterns in market microstructure noise. These results can shed more light on the sign of noise autocorrelation in the presence of market microstructure frictions such as bid-ask bounces and the clustering of order flow. (*JEL*: C12, D49)

**KEYWORDS:** high-frequency data, market microstructure noise, nonsynchronous observations, test statistic, time dependence

Market microstructure noise can be induced by various market frictions such as the discreteness of price changes, bid-ask bounces, and asymmetric information across traders, *inter alia*. The growing literature on market microstructure provides important insights from early studies including Roll (1984), who derives a simple estimator of bid-ask spread based on the negative autocovariance of returns. The covariance spread model proposed by Stoll (1989) and George, Kaul, and Nimalendran (1991) provides an alternative approach to the examination of bid-ask spread. The early literature has also focused on statistical inferences about bid-ask spread decomposition using an indicator function, which is driven by trade direction as in Glosten and Harris (1988), Huang and Stoll (1997), and Madhavan,

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Richardson, and Roomans (1997). In contrast, Harris (1990) examines the rounding effects emanating from the discreteness of transaction prices. There are studies suggesting that asymmetric information across traders causes adverse selection components within bid-ask spreads in both the market-maker system and limit-order markets (see, for instance, Glosten and Milgrom 1985; Easley and O'Hara 1987; Copeland and Galai 1983; Foucault 1999; and Glosten 1994, *inter alia*).

The recent literature on microstructure noise provides new evidence on the important issues of whether and to what extent microstructure noise causes transaction prices to deviate from frictionless equilibrium prices. Bandi and Russell (2006) propose a new approach to the assessment of full-information transaction costs, which represent the difference between transaction prices and unobserved theoretical asset prices reflecting all private and public information. Also, Meddahi (2002) and Hansen and Lunde (2006) examine the variance of noise as well as the correlation between noise and the integrated variance of semimartingale price process in frictionless markets.

The literature on integrated variance estimation with noise includes the earlier study by Zhou (1996), who proposes a kernel-based estimator. In contrast, Zhang, Mykland, and Aït-Sahalia (2005) suggest two scales of realized variance as a linear combination of realized variances at two frequencies, with Zhang (2006) extending this estimator to multiple scales. It is noted that these studies are based on the underlying assumption of independently and identically distributed noise. It is possible, however, that market microstructure noise is governed by time dependence. Under the more plausible assumption of noise dependence, Aït-Sahalia, Mykland, and Zhang (2006) propose new estimators of two- and multiple-scale realized variances, while Hansen and Lunde (2005, 2006) and Barndorff-Nielsen et al. (2008) develop the kernel-based estimator. Also, the optimal frequency based on the minimization of the mean squared errors is proposed by Bandi and Russell (2008).

As far as the estimation of integrated covariance under noise dependence is concerned, it is noted that the cumulative covariance estimator proposed by Hayashi and Yoshida (2005) is shown to be biased in the presence of cross-correlated noise. Voev and Lunde (2007) propose instead modified cumulative covariance estimators based on subsampling and kernel approaches, in which a new test statistic for noise cross-sectional dependence is proposed in order to determine the kernel bandwidth. The present study differs from Voev and Lunde (2007) in many respects. These differences pertain to the choice of intervals used for the estimation of noise cross-covariance and an evaluation of a variance of the noise cross-covariance estimator. First, it is noted that the cross-covariance estimation by Voev and Lunde (2007) is based on subsampling intervals, which are large enough to ensure unbiasedness. However, wider subsampling intervals may be conducive to a larger variance of the noise cross-covariance estimator. To eschew these problems, a new approach to noise cross-covariance estimation is proposed based on the interval length for which noise dependence disappears. The testing procedure used to determine the appropriate length of the sampling interval is also described in this study. Second, in order to assess the unbiasedness of

kernel-based cumulative covariance estimators, Voev and Lunde (2007) propose  $t$ -statistics based on an approximated variance of the noise cross-covariance estimator under i.i.d. noise. As noted in their paper, it is natural that this approximation leads to somewhat larger  $t$ -statistics than they should be. In contrast, the test statistics derived in this paper do not require the i.i.d. noise approximation using subsampling methods.

These estimation issues are important in the sense that the statistical properties in terms of unbiasedness, consistency, and efficiency can be intrinsically linked to the noise dependence structures. Indeed, many empirical studies provide evidence of significant autocorrelations and lead-lag relationship of cross-correlation in intraday returns. Arguably, this correlation can be induced, however, by the dependence of microstructure noise. A better understanding of time dependence in market microstructure noise is also warranted insofar as it renders possible the bias correction in the estimation of integrated variance and covariance.

It is, thus, the main purpose of this study to examine the degree of dependence in microstructure noise. The main contribution of this study lies in the construction of new test statistics for dependence and covariance estimators of bivariate noise processes. The related asymptotic distributions are also derived. Furthermore, test statistics for significance of the noise variance and covariance are provided. It should be noted, however, from the outset, that the focus is not made so much on equilibrium price covariance estimation as on the statistical properties of noise dependence. This focus on noise dependence, which has the potential of contaminating high-frequency prices, is also warranted because it has significant implications for option pricing, value-at risk analysis, and portfolio management. The evidence from Monte Carlo simulation indicates that the proposed cross and auto covariance estimators and test statistics perform better in finite samples. The empirical evidence from high-frequency data, for a sample of Japanese stocks traded on the Osaka Securities Exchange, is suggestive of varying patterns of dependence in market microstructure noise.

The remainder of the paper is organized as follows. In Section 1, the transaction price model and the properties of market microstructure noise are discussed within the framework of high-frequency financial data analysis. The new test statistic for cross-sectional dependence in the noise process is described in Section 2. The cross and auto covariance estimators of the bivariate noise process, as well as their asymptotic distributions, and tests of significance are examined in Section 3. Section 4 presents the results of Monte Carlo simulation and the empirical illustration from high-frequency data. Section 5 concludes the paper with an Appendix that provides proofs for the lemmas and theorems.

## 1 PRICE PROCESS AND MARKET MICROSTRUCTURE NOISE

It is assumed that the logarithmic equilibrium price processes of two assets,  $\{P_1^*(t)\}$  and  $\{P_2^*(t)\}$ , follow a two-dimensional Itô process without drift on a probability

space  $(\Omega, \mathcal{F}, \mathcal{P})$ :

$$\begin{aligned} dP_y^*(t) &= \sigma_y(t) dW_y(t), \quad y = 1, 2, \quad t \in [0, T], \\ d\langle W_1, W_2 \rangle_t &= \rho^*(t) dt, \quad \rho^*(t) \in (-1, 1), \end{aligned} \quad (1)$$

where  $W_1$  and  $W_2$  are standard Brownian motions. The initial value of the price  $P_y^*(0)$  is a constant and  $\sigma_y(t) > 0$  is a bounded progressively measurable function.  $\{P_1^*(t)\}$  and  $\{P_2^*(t)\}$  represent the logarithmic market prices without trading imperfections, market frictions, and informational effects.

Because of irregularly nonsynchronous trading, the  $i$ th observed intraday return on asset 1 is denoted by  $r_{1,i} := P_1(t_i) - P_1(t_{i-1})$  whereas  $r_{2,j} := P_2(s_j) - P_2(s_{j-1})$  represents the  $j$ th return on asset 2, where  $t_i$  and  $s_j$  are the end times of the  $i$ th and  $j$ th intervals. The hidden semimartingale process named by Mykland and Zhang (2005) is used in modeling high-frequency transaction prices. The logarithmic transaction price  $P_y$  with market microstructure noise is described as follows:

$$P_1(t_i) = P_1^*(t_i) + \eta(t_i), \quad P_2(s_j) = P_2^*(s_j) + \delta(s_j), \quad (2)$$

where  $P_y^*(t)$  is the logarithmic equilibrium price defined in (1). The market microstructure noise, which may be due to bid-ask spreads, the discreteness of prices change and the difference in trade sizes, is represented by  $\eta(t_i)$  and  $\delta(s_j)$  for asset 1 and 2, respectively. It is assumed that market microstructure noise has the following properties.

**Assumption 1.** *Market microstructure noise.*

Let a vector of market microstructure noise of asset 1 and 2 be represented by  $\mathbf{u}(t) = (\eta(t) \delta(t))'$ .

(1a)  $\{\mathbf{u}(t)\}$  is a sequence of random variables with zero mean.

(1b) The bivariate noise processes are covariance stationary, with the autocovariance function having a finite dependence in the sense that:

$$\Gamma(\ell) = E[\mathbf{u}(t)\mathbf{u}'(t - \ell)] = \begin{pmatrix} \gamma_\eta(\ell) & \gamma_{\eta\delta}(\ell) \\ \gamma_{\delta\eta}(\ell) & \gamma_\delta(\ell) \end{pmatrix} = \mathbf{0}, \quad \text{for all } |\ell| > m,$$

where  $m$  is a finite positive number.  $\Gamma(\ell)$  is a covariance matrix with finite elements.

(1c) There exists some positive number  $\beta > 1$  that satisfies  $E|\mathbf{u}(t)\mathbf{u}'(s)|^{4\beta} < \infty$  for all  $t, s$ .

(1d) The noise process is independent of the equilibrium price process.  $P_y^* \perp \mathbf{u}(t)$ ,  $y = 1, 2$ .

For the sake of simplicity and given that  $\gamma_{\eta\delta}(\ell) = E[\eta(t)\delta(t - \ell)]$  and  $\gamma_{\delta\eta}(\ell) = E[\eta(t - \ell)\delta(t)] = E[\eta(t)\delta(t + \ell)]$ , the subscripts in (1b) are dropped and  $\gamma_{\eta\delta}(\ell)$  and  $\gamma_{\delta\eta}(\ell)$  are rewritten as  $\gamma(\ell)$  and  $\gamma(-\ell)$ . The auto and cross correlation coefficients of the two noise terms are denoted by  $\rho_\eta(\ell)$ ,  $\rho_\delta(\ell)$  and  $\rho(\ell)$ , respectively. As the number of high-frequency observations increases, the dependence between the noise terms generally dominates that between the equilibrium price and noise even if  $P_y^*$  and

$u(t)$  are correlated in (1d). Furthermore, Hansen and Lunde (2006) suggest that the independence assumption between the equilibrium price and noise does not statistically damage the analysis of asset prices with high trading intensities.

The focus in the present study is centered upon the estimation of the covariance matrix, which is defined in Assumption (1b). The asymptotic and finite sample properties of the auto and cross covariance estimators of bivariate noise processes are characterized. Furthermore, a test statistic for the cross-sectional dependence of the noise terms using the subsampling method is introduced hereafter.

## 2 CROSS-SECTIONAL DEPENDENCE OF NOISES

It is important to note that the microstructure noise as well as equilibrium prices are unobservable. Thus, a simple identification approach is introduced in order to measure the cross-sectional dependence of noise. A test statistic is proposed to identify the distance with respect to which the noise dependence disappears.

### 2.1 Measurement of Cross-Sectional Noise Dependence

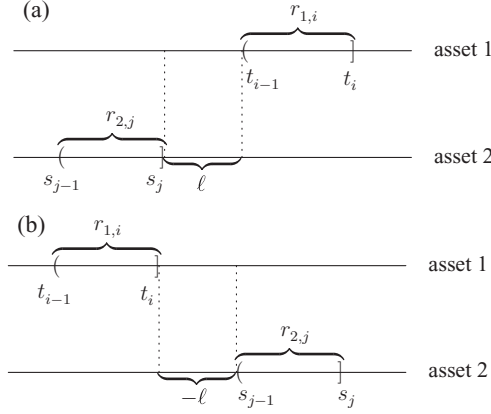
Given intervals  $I^i := (t_{i-1}, t_i]$  and  $J^j := (s_{j-1}, s_j]$  for all  $i, j$ , the expectation and variance conditional on the stochastic arrival times are defined as  $E_{IJ}[\cdot]$  and  $V_{IJ}[\cdot]$ . It is not possible to identify the covariation of the equilibrium price processes that have martingale properties and the covariance of market microstructure noise with respect to the pair of overlapping intervals because the covariation of equilibrium prices is different from zero. On the other hand, for the pair of nonoverlapping intervals  $\{I^i \cap J^j = \emptyset\}$ :

$$E_{IJ} \left[ \sum_{i,j} r_{1,i} r_{2,j} 1_{\{I^i \cap J^j = \emptyset\}} \right] = E_{IJ} \left[ \sum_{i,j} e_{\eta,i} e_{\delta,j} 1_{\{I^i \cap J^j = \emptyset\}} \right], \quad (3)$$

where  $e_{\eta,i} := \eta(t_i) - \eta(t_{i-1})$  and  $e_{\delta,j} := \delta(s_j) - \delta(s_{j-1})$ . The product of returns on nonoverlapping intervals is used to identify the covariance of noises as in Voev and Lunde (2007).

For the nonoverlapping intervals  $\{I^i \cap J^j = \emptyset\}$  and  $t_{i-1} - s_j > 0$ , the distance between intervals is defined as  $\ell = t_{i-1} - s_j$  (Figure 1, panel a). In the case where  $s_{j-1} - t_i > 0$ , the difference of the nonoverlapping intervals is denoted by  $\ell = -(s_{j-1} - t_i)$ , (Figure 1, panel b). The nonoverlapping adjacent intervals such that  $t_{i-1} - s_j = 0$  or  $s_{j-1} - t_i = 0$  are used in the case where  $\ell = 0$ . In what follows, we consider the case of  $\ell > 0$  because, for the other cases, we have only to replace the corresponding definition of  $\ell$ . It is noted that  $\ell = t_{i-1} - s_j > 0$  implies  $\{I^i \cap J^j = \emptyset\}$ . The product of returns on the  $i$ th and  $j$ th intervals such that  $t_{i-1} - s_j = \ell > 0$  is defined as follows:

$$Z_{\ell,ij} = r_{1,i} r_{2,j} \text{ for all } i, j, \text{ such that } t_{i-1} - s_j = \ell. \quad (4)$$



**Figure 1** Pairs of returns on nonoverlapping intervals with (a)  $\ell > 0$  and (b)  $\ell < 0$ .

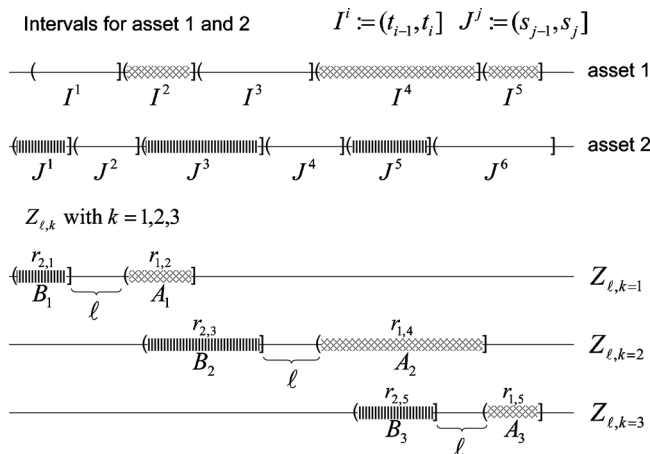
The conditional expectation of  $Z_{\ell,ij}$  is expressed as

$$\begin{aligned} E_{\Pi}[Z_{\ell,ij}] &= E_{\Pi}[\eta(t_i)\delta(s_j)] - E_{\Pi}[\eta(t_i)\delta(s_{j-1})] - E_{\Pi}[\eta(t_{i-1})\delta(s_j)] + E_{\Pi}[\eta(t_{i-1})\delta(s_{j-1})] \\ &= \gamma(\ell + \Delta t_i) - \gamma(\ell + \Delta t_i + \Delta s_j) - \gamma(\ell) + \gamma(\ell + \Delta s_j), \end{aligned} \quad (5)$$

where  $\Delta t_i := t_i - t_{i-1}$  and  $\Delta s_j := s_j - s_{j-1}$ . For all  $\ell$  exceeding a large enough value  $L$  such that  $\gamma(L) = 0$ , it follows that  $\gamma(\ell) = 0$  and  $E_{\Pi}[Z_{\ell,ij}] = 0$  from (1b) in Assumption 1. Now suppose that  $s^* := \min_s \{s \mid \gamma(L - s) \neq 0, s \geq 0\}$ . This implies that  $\gamma(L) = \gamma(L - 1) = \gamma(L - 2) = \dots = \gamma(L - s^* + 1) = 0$  and  $\gamma(L - s^*) \neq 0$ , and  $E_{\Pi}[Z_{L,ij}] = E_{\Pi}[Z_{L-1,ij}] = E_{\Pi}[Z_{L-2,ij}] = \dots = E_{\Pi}[Z_{L-s^*+1,ij}] = 0$  and  $E_{\Pi}[Z_{L-s^*,ij}] \neq 0$  because  $\gamma(L - s^*) \neq 0$ . It results from  $\ell^* = L - s^*$ , that  $\gamma(\ell^* + 1) = 0$  and  $\gamma(\ell^*) \neq 0$ , which implies that the distance  $\ell^*$  represents the threshold value of noise dependence. It is important to note that  $E_{\Pi}[Z_{\ell,ij}] = 0$  does not necessarily imply that  $\gamma(\ell) = 0$ . This is due to the fact that the sum of all cross-covariances in (5) can still take the value of zero, even when  $\gamma(\ell) \neq 0$ . To avoid such restrictions, the method of determination for the threshold value  $\ell^*$  described above is applied. Thus, a test statistic for determining the threshold value can be constructed using the sample mean of  $Z_{\ell,ij}$ , which satisfies the nonoverlapping intervals with distance  $\ell$ .

## 2.2 Test Statistic for Cross-Sectional Dependence of Noises

The intuition behind Assumption (1b) is that the dependence between two noises disappears when the noise terms are sufficiently separated. As in the previous subsection, a new test statistic is hereafter proposed to identify the threshold value. The observations  $Z_{\ell,ij}$  are arranged in ascending order of index  $i$  as  $\{Z_{\ell,k}\}_{k=1}^{N_{\ell}}$ , where  $N_{\ell}$  is the total number of the products of returns on the nonoverlapping intervals with distance  $\ell$ . In the resulting sequence  $\{Z_{\ell,k}\}_{k=1}^{N_{\ell}}$ , the  $k$ th pair of the selected



**Figure 2** An example of each pair of the intervals  $(A_k, B_k)$  for  $k = 1, 2, 3$ .

intervals are denoted as  $A_k$  and  $B_k$ . The product of returns on nonoverlapping intervals  $A_k$  and  $B_k$  is defined as  $Z_{\ell,k}$ . Figure 2 illustrates each pair of intervals  $(A_k, B_k)$ ,  $(A_{k+1}, B_{k+1})$ , and  $(A_{k+2}, B_{k+2})$  for  $k = 1$ . The sample mean  $Z_{\ell,k}$ , that is,  $\bar{Z}_{\ell, N_\ell} := \frac{1}{N_\ell} \sum_{k=1}^{N_\ell} Z_{\ell,k}$ , is used to estimate  $E_{\Pi}[Z_{\ell,k}]$ . Let  $f_{\ell, N_\ell} := (\bar{Z}_{\ell, N_\ell} - E_{\Pi}[Z_{\ell, N_\ell}])N_\ell^{1/2}$  be the theoretical standardization for  $\bar{Z}_{\ell, N_\ell}$ . The following assumption is made with respect to the asymptotic variance of  $f_{\ell, N_\ell}$ , which is given by  $\lim_{N_\ell \rightarrow \infty} E_{\Pi}[(f_{\ell, N_\ell})^2] = \sigma_{\ell, f}^2$ :

**Assumption 2.**  $V_{\Pi}[n^{-1/2} \sum_{k=k'+1}^{k'+n} Z_{\ell,k}] \rightarrow \sigma_{\ell, f}^2$ , uniformly in any  $k'$ , as  $n \rightarrow \infty$ . This means that  $\sup_{k'} |V_{\Pi}[n^{-1/2} \sum_{k=k'+1}^{k'+n_{N_\ell}} Z_{\ell,k}] - \sigma_{\ell, f}^2| \rightarrow 0$  as  $N_\ell \rightarrow \infty$  for any sequence  $\{n_{N_\ell}\}$  that tends to infinity.

With respect to the asymptotic normality of the estimator of  $E_{\Pi}[Z_{\ell,k}]$ , it is noted that  $\{Z_{\ell,k}\}_{k=1}^{N_\ell}$  is a sequence of dependent and heterogeneously distributed random scalars because the variance depends on the length of the irregularly observed intervals and  $\{Z_{\ell,k}\}$  is serially correlated. Thus, the following lemma applies for the asymptotic normality of the estimator of  $E_{\Pi}[Z_{\ell,k}]$ .

**Lemma 1.** Suppose Assumptions 1 and 2 hold. As  $N_\ell$  goes to infinity, we have:

$$\frac{f_{\ell, N_\ell}}{\sigma_{\ell, f}} \xrightarrow{a} N(0, 1). \quad (6)$$

The proof is given in the Appendix.

Although the asymptotic variance  $\sigma_{\ell, f}^2$  is unknown, it is possible to construct a consistent estimator of  $\sigma_{\ell, f}^2$  by applying the subsampling method proposed by

Carlstein (1986), who considers variance estimation for a general statistic without specifying the dependence in a stationary sequence. Fukuchi (1999) and Politis, Romano, and Wolf (1999) extend the results to heteroskedastic observations.

The total sample is divided into nonoverlapping subseries that consist of  $M_\ell$  consecutive observations. Thus, the  $h$ th subseries using  $\{Z_{\ell,k}\}_{k=1}^{N_\ell}$  can be defined as follows:

$$\{Z_\ell^h\} := (Z_{\ell,hM_\ell+1}, Z_{\ell,hM_\ell+2}, \dots, Z_{\ell,(h+1)M_\ell}), \quad 0 \leq h \leq K_\ell - 1, \quad K_\ell = [N_\ell/M_\ell],$$

where  $[\cdot]$  denotes the integer part of a real number,  $M_\ell$  is a number of observations within each subseries  $\{Z_\ell^h\}$ , and  $K_\ell$  denotes the total number of subseries. The superscript  $h$  of  $\{Z_\ell^h\}$  implies that the subseries is associated with the sample  $Z_{\ell,k}$  with  $k = hM_\ell + 1, \dots, (h+1)M_\ell$ . The variance estimator can be expressed as

$$\hat{\sigma}_{\ell,f}^2 = \frac{M_\ell}{K_\ell} \sum_{h=0}^{K_\ell-1} \left( \bar{Z}_{\ell,M_\ell}^h - \frac{1}{K_\ell} \sum_{h=0}^{K_\ell-1} \bar{Z}_{\ell,M_\ell}^h \right)^2, \quad (7)$$

where  $\bar{Z}_{\ell,M_\ell}^h$  is the sample mean of subseries  $\{Z_\ell^h\}$ . The following lemma applies with respect to the variance estimator  $\hat{\sigma}_{\ell,f}^2$ .

**Lemma 2.** *Suppose Assumptions 1 and 2 hold. Let  $M_\ell$  be s.t.  $M_\ell \rightarrow \infty$  and  $M_\ell/N_\ell \rightarrow 0$  as  $N_\ell \rightarrow \infty$ . Then we have*

$$\hat{\sigma}_{\ell,f}^2 \rightarrow_{L_2} \sigma_{\ell,f}^2 \quad \text{as } N_\ell \rightarrow \infty. \quad (8)$$

The proof is provided in the Appendix. It is noted that the other subsampling estimation using overlapping subseries is also applicable (see Politis, Romano, and Wolf (1999) for details).

The test statistic that identifies the threshold value of noise dependence is derived from the results of Lemmas 1 and 2. Let the null hypothesis be  $E_{ij}[Z_{\ell,k}] = 0$  for all  $k$ , for given  $\ell$  while the alternative hypothesis consists of all possible deviations from the null. Then, the following theorem can be derived for the new test statistic.

**Theorem 1.** *Suppose Assumptions 1 and 2 hold. As  $N_\ell$  goes to infinity s.t.  $M_\ell \rightarrow \infty$  and  $M_\ell/N_\ell \rightarrow 0$ , we have*

$$\tau(\ell) := \frac{\sqrt{N_\ell} \bar{Z}_{\ell,N_\ell}}{\hat{\sigma}_{\ell,f}} \xrightarrow{a} N(0, 1) \quad (9)$$

under the null hypothesis.  $\tau(\ell)$  diverges under the alternative.  $\hat{\sigma}_{\ell,f}$  is the square root of variance estimator given by (7).

The asymptotic normality of the test statistic follows directly from Lemmas 1 and 2. The existence of large numbers  $M_\ell$  and  $K_\ell$  for the variance estimation



in (7) is reflective of the fact that high-frequency transaction data yields a large number  $N_\ell$ . However, it is difficult to determine the optimal values of  $M_\ell$  and  $K_\ell$  that minimize the mean squared error of  $\hat{\sigma}_{\ell,f}^2$  because the noise covariance structure remains unknown a priori. For a fixed number  $N_\ell$ , an increase in  $M_\ell$  (i.e., a decrease in  $K_\ell$ ) reduces the bias but increases the variance of  $\hat{\sigma}_{\ell,f}^2$ . It is known that the optimal asymptotic rate of  $M_\ell$  is proportional to  $N_\ell^{1/3}$  for the subsampling variance estimation given that  $M_\ell = cN_\ell^{1/3}$  where  $c$  depends on the noise process. Prior to the examination of  $M_\ell$  and  $K_\ell$  selected under  $M_\ell = cN_\ell^{1/3}$  with some  $c$  through Monte Carlo simulation, it is important to describe in further detail, the procedure for identifying threshold values for which the noise cross-covariance is reduced to zero. First, it is noted that the null hypothesis  $E_{\Pi}[Z_{L,k}] = 0$  for all  $k$  with a large value  $L$  using the test statistic (9) may not be rejected because it follows from (1b) of Assumption 1 that the cross-covariance between sufficiently separated noise terms is zero. When  $E_{\Pi}[Z_{L,k}] = 0$  is not rejected as expected, then it is necessary to test whether  $E_{\Pi}[Z_{L-1,k}]$  is zero. Again, if  $E_{\Pi}[Z_{L-1,k}] = 0$  is not rejected, then the statistical significance of  $E_{\Pi}[Z_{L-2,k}]$  should be assessed. This test is reiterated sequentially until the null is rejected. Finally, the distance for which the null is rejected for the first time can be determined as  $\ell^* = \max_\ell \{|\tau(\ell)| > c.v.\}$ , where  $c.v.$  is the critical value of the test statistic (9).

### 3 CONSISTENT NOISE COVARIANCE ESTIMATORS

The cross-covariance of bivariate noise processes  $\gamma(\ell)$  for  $\ell > \ell^*$  is zero when the threshold value  $\ell^*$  is selected as described in Section 2. However,  $\gamma(\ell)$  is still unknown for  $\ell < \ell^*$ . The cross-covariance estimator of bivariate noise processes and its asymptotic distribution will be derived in Section 3.1, where a test statistic for statistical significance of the cross-covariance is also described. The related statistics for autocovariance of univariate noise process are presented in Section 3.2.

#### 3.1 Noise Cross-Covariance Estimation

For the purposes of constructing an unbiased estimator, it is necessary to identify the nonoverlapping intervals for which all cross-covariances in (5) except  $\gamma(\ell)$  become zero. This interval depends on the threshold value, which is determined using the test statistic (9). The threshold values  $m^+$  and  $m^-$  are used instead of  $\ell^*$  in case where  $\ell > 0$  and  $\ell < 0$ , respectively. Suppose that the bivariate noise processes have finite cross-sectional dependence in the sense that  $\gamma(\ell) = 0$  for  $\ell > m^+ > 0$  and for  $-\ell > m^- > 0$ . When  $\ell$  is positive, it is possible to define  $\bar{t}_i^{(+)}$  as the first transaction time for asset 1, which follows  $t_i$  subject to  $\bar{t}_i^{(+)} - s_j > m^+$ . Likewise,  $\underline{s}_{j-1}^{(+)}$  can be defined as the last transaction time for asset 2, which is followed by  $s_{j-1}$  subject to  $t_{i-1} - \underline{s}_{j-1}^{(+)} > m^+$ . As  $\ell$  is negative,  $\bar{s}_j^{(-)}$  refers to the first transaction time for asset 2, which follows  $s_j$  subject to  $\bar{s}_j^{(-)} - t_i > m^-$  while  $\underline{t}_{i-1}^{(-)}$  is the last transaction time for asset 1, which is followed by  $t_{i-1}$  subject to  $s_{j-1} - \underline{t}_{i-1}^{(-)} > m^-$ . The returns

on the intervals  $(t_{i-1}, \bar{t}_i^{(+)})$  and  $(\underline{t}_{i-1}^{(-)}, t_i]$  are denoted by  $\bar{r}_{1,i}^{(+)} := P_1(\bar{t}_i^{(+)}) - P_1(t_{i-1})$  and  $\underline{r}_{1,i}^{(-)} := P_1(t_i) - P_1(\underline{t}_{i-1}^{(-)})$ , respectively. Similarly, for asset 2, the returns on the intervals  $(s_{j-1}, \bar{s}_j^{(+)})$  and  $(\underline{s}_{j-1}^{(-)}, s_j]$  are denoted by  $\bar{r}_{2,j}^{(+)} := P_2(\bar{s}_j^{(+)}) - P_2(s_{j-1})$  and  $\underline{r}_{2,j}^{(-)} := P_2(s_j) - P_2(\underline{s}_{j-1}^{(-)})$ , respectively. It is possible thus to define  $Z_{\ell,ij}^{(\pm)}$ , in relation to  $Z_{\ell,ij}$  in (4), as follows:

$$Z_{\ell,ij}^{(\pm)} = \begin{cases} \bar{r}_{1,i}^{(+)} \underline{r}_{2,j}^{(+)} \mathbf{1}_{\{t_{i-1}-s_j=\ell\}} & \text{if } \ell > 0 \\ \bar{r}_{1,i}^{(+)} \underline{r}_{2,j}^{(+)} \mathbf{1}_{\{t_{i-1}-s_j=0\}} + \underline{r}_{1,i}^{(-)} \bar{r}_{2,j}^{(-)} \mathbf{1}_{\{s_{j-1}-t_i=0\}} & \text{if } \ell = 0 \\ \underline{r}_{1,i}^{(-)} \bar{r}_{2,j}^{(-)} \mathbf{1}_{\{s_{j-1}-t_i=-\ell\}} & \text{if } \ell < 0. \end{cases} \quad (10)$$

Panels (a), (b), and (c) of Figure 3 illustrate each pair of intervals with respect to  $\ell > 0$ ,  $\ell = 0$ , and  $\ell < 0$ , respectively. For all  $\ell > 0$ , the cross-covariances are zero when the distance between noise terms exceeds  $m^+$ . The nonoverlapping intervals described in panels (b) and (c) are given by the same idea as in panel (a). The conditional expectation of  $Z_{\ell,ij}^{(\pm)}$ , where the indicator function takes the value of one, is denoted as  $E_{\Pi}[Z_{\ell,ij}^{(\pm)}] = -\gamma(\ell)$ . We select  $Z_{\ell,ij}^{(\pm)}$  for all  $i, j$  such that the indicator function is equal to one. The ranking of the selected  $Z_{\ell,ij}^{(\pm)}$  in ascending order of index  $i$  results in the sequence  $\{Z_{\ell,k}^{(\pm)}\}_{k=1}^{N_{\ell}^{(\pm)}}$ . The sample size  $N_{\ell}^{(\pm)}$  is not greater than  $N_{\ell}$ , which is the sample size of  $\{Z_{\ell,k}\}$  described in Section 2.2. The cross-covariance estimator can thus be calculated as

$$\hat{\gamma}(\ell) = -\frac{1}{N_{\ell}^{(\pm)}} \sum_{k=1}^{N_{\ell}^{(\pm)}} Z_{\ell,k}^{(\pm)}. \quad (11)$$

The following theorem describes the asymptotic normality of cross-covariance estimator  $\hat{\gamma}(\ell)$ .

**Theorem 2.** Suppose Assumptions 1 and 2 hold. Let  $M_{\ell}$  be s.t.  $M_{\ell} \rightarrow \infty$  and  $M_{\ell}/N_{\ell}^{(\pm)} \rightarrow 0$  as  $N_{\ell}^{(\pm)} \rightarrow \infty$ . Given the threshold values  $m^+$  and  $m^-$  determined through the test statistic (9), we have

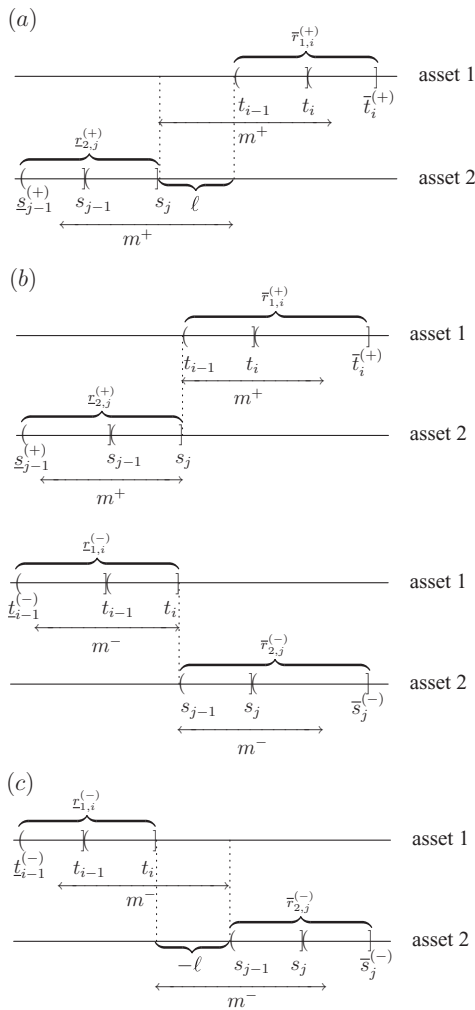
$$\sqrt{N_{\ell}^{(\pm)}}(\hat{\gamma}(\ell) - \gamma(\ell)) \xrightarrow{a} N(0, \omega_{\ell}^2), \quad (12)$$

where

$$\omega_{\ell}^2 = \lim_{N_{\ell}^{(\pm)} \rightarrow \infty} E_{\Pi}[N_{\ell}^{(\pm)}(\hat{\gamma}(\ell) - \gamma(\ell))^2].$$

The proof is provided in the Appendix. It is noted that the conditions for the threshold value selection subject to  $M_{\ell} \rightarrow \infty$  and  $M_{\ell}/N_{\ell} \rightarrow 0$  as  $N_{\ell} \rightarrow \infty$  hold since we have  $N_{\ell}^{(\pm)} \leq N_{\ell}$ .

For the sake of the notational simplicity, we will denote the sample size of  $\{Z_{\ell,k}^{(\pm)}\}$  by  $N_{\ell}$  instead of  $N_{\ell}^{(\pm)}$ . The subsampling variance estimator of  $\omega_{\ell}^2$  based



**Figure 3** Pairs of returns on extended intervals in case of (a)  $\ell > 0$ , (b)  $\ell = 0$  and (c)  $\ell < 0$ .

on nonoverlapping subseries, which has  $M_\ell$  consecutive observations, can be expressed as

$$\hat{\omega}_\ell^2 = \frac{M_\ell}{K_\ell} \sum_{h=0}^{K_\ell-1} \left( \bar{Z}_{\ell, M_\ell}^{(\pm), h} - \frac{1}{K_\ell} \sum_{h=0}^{K_\ell-1} \bar{Z}_{\ell, M_\ell}^{(\pm), h} \right)^2, \quad (13)$$

where  $\bar{Z}_{\ell, M_\ell}^{(\pm), h}$  is the sample mean of the subseries  $\{Z_\ell^{(\pm), h}\}$ . The  $h$ th subseries is defined as

$$\{Z_\ell^{(\pm), h}\} := (Z_{\ell, hM_\ell+1}^{(\pm)}, Z_{\ell, hM_\ell+2}^{(\pm)}, \dots, Z_{\ell, (h+1)M_\ell}^{(\pm)}), \quad 0 \leq h \leq K_\ell - 1, \quad K_\ell = \lceil N_\ell / M_\ell \rceil.$$

Thus, the asymptotic distribution of  $\hat{\gamma}(\ell)$  described in Theorem 2 and the subsampling variance estimator can be used in the test statistic of the null hypothesis  $\gamma(\ell) = 0$  against the alternative  $\gamma(\ell) \neq 0$ .

**Corollary 1.** *The test for the cross-covariance of market microstructure noises.*

*Suppose Assumptions 1 and 2 hold. As  $N_\ell$  goes to infinity s.t.  $M_\ell \rightarrow \infty$  and  $M_\ell/N_\ell \rightarrow 0$ , we have:*

$$\tau^*(\ell) := \frac{\sqrt{N_\ell} \hat{\gamma}(\ell)}{\hat{\omega}_\ell} \xrightarrow{a} N(0, 1) \quad (14)$$

*under the null hypothesis.  $\tau^*(\ell)$  diverges under the alternative.  $\hat{\omega}_\ell$  is the square root of variance estimator given by (13).*

It is noted that in the estimation of the integrated covariance that the bias of cumulative covariance estimator suggested by Hayashi and Yoshida (2005) is virtually zero when the test statistic (14) does not reject the null  $\gamma(\ell) = 0$  for all  $\ell$ . However, even if the cumulative covariance estimator is unbiased, the noise terms can still affect the variance. Thus, applying the subsampling method to derive the cumulative covariance estimator by Hayashi and Yoshida (2005) can be still preferable for such cases. This approach is also consistent with the subsampling integrated covariance estimation by Voev and Lunde (2007).

### 3.2 Autocovariance Estimator of Noise

The approach to the estimation of noise cross-covariance and the derivation of test statistic of significance described in the previous subsection applies with equal force to the autocovariance of the univariate noise process. The estimation of a consistent noise autocovariance requires the application of test statistic to determine the threshold value for univariate noise dependence. The product of returns on the  $i$ th and  $j$ th intervals for asset 1 such that  $\ell = t_{j-1} - t_i \geq 0$  is defined as follows:

$$Z_{1,\ell,ij} = r_{1,i}r_{1,j} \quad \text{for all } i, j, \quad \text{such that } t_{j-1} - t_i = \ell \geq 0. \quad (15)$$

Let  $\{Z_{1,\ell,k}\}_{k=1}^{N_{1,\ell}}$  be the sequence of  $Z_{1,\ell,ij}$ , which is arranged in ascending order of the index  $i$ . The total number of observations in the sequence  $\{Z_{1,\ell,k}\}$  is denoted as  $N_{1,\ell}$ . For a given  $\ell$ , the null hypothesis is  $E_{\Pi}[Z_{1,\ell,k}] = 0$  for all  $k$ , which implies that  $\gamma_\eta(\ell) = 0$ . The alternative hypothesis consists of all possible deviations from the null. The test statistic for the null is expressed as

$$\tau_\eta(\ell) := \frac{\sqrt{N_{1,\ell}} \bar{Z}_{1,\ell,N_{1,\ell}}}{\hat{\sigma}_{1,\ell,f}}. \quad (16)$$

$\hat{\sigma}_{1,\ell,f}^2$  is a subsampling estimator of  $\sigma_{1,\ell,f}^2 = \lim_{N_{1,\ell} \rightarrow \infty} E_{\Pi}[(f_{1,\ell,N_{1,\ell}})^2]$ , in which  $f_{1,\ell,N_{1,\ell}} := (\bar{Z}_{1,\ell,N_{1,\ell}} - E_{\Pi}[\bar{Z}_{1,\ell,N_{1,\ell}}])N_{1,\ell}^{1/2}$ . The sample mean of  $\{Z_{1,\ell,k}\}$  is denoted as

$\bar{Z}_{1,\ell,N_{1,\ell}}$  and the subsampling variance estimator can be expressed as

$$\hat{\sigma}_{1,\ell,f}^2 = \frac{M_{1,\ell}}{K_{1,\ell}} \sum_{h=0}^{K_{1,\ell}-1} \left( \bar{Z}_{1,\ell,M_{1,\ell}}^h - \frac{1}{K_{1,\ell}} \sum_{h=0}^{K_{1,\ell}-1} \bar{Z}_{1,\ell,M_{1,\ell}}^h \right)^2, \quad (17)$$

where  $\bar{Z}_{1,\ell,M_{1,\ell}}^h$  is the sample mean of the  $h$ th subseries  $\{Z_{1,\ell}^h\}$ :

$$\{Z_{1,\ell}^h\} := (Z_{1,\ell,hM_{1,\ell}+1}, Z_{1,\ell,hM_{1,\ell}+2}, \dots, Z_{1,\ell,(h+1)M_{1,\ell}}), \\ 0 \leq h \leq K_{1,\ell}-1, \quad K_{1,\ell} = [N_{1,\ell}/M_{1,\ell}].$$

Under the null, it is clear that  $\tau_\eta(\ell) \xrightarrow{a} N(0, 1)$  as  $N_{1,\ell}$  goes to infinity s.t.  $M_{1,\ell} \rightarrow \infty$  and  $M_{1,\ell}/N_{1,\ell} \rightarrow 0$ . The threshold value  $m_1$  of the finite noise dependence for asset 1 implies that the autocovariance function  $\gamma_\eta(\ell)$  for all  $\ell > m_1$  is zero. The test statistic (16) allows for the identification of the threshold value  $m_1$ .

In order to derive the noise autocovariance estimator, the sequence  $Z_{1,\ell,ij}^{(\pm)}$  for all  $i, j$  such that  $t_{j-1} - t_i = \ell \geq 0$  is constructed using the selected threshold value  $m_1$  as follows:

$$Z_{1,\ell,ij}^{(\pm)} = \bar{r}_{1,i}^{(-)} \bar{r}_{1,j}^{(+)} = \left( P(t_i) - P(\bar{t}_{i-1}^{(-)}) \right) \left( P(\bar{t}_j^{(+)} - P(t_{j-1})) \right), \\ \text{for all } i, j, \text{ such that } t_{j-1} - t_i = \ell \geq 0, \quad (18)$$

where  $\bar{t}_j^{(+)}$  is the first transaction time, which immediately follows  $t_j$  subject to  $\bar{t}_j^{(+)} - t_i > m_1$ , and  $\bar{t}_{i-1}^{(-)}$  is the last transaction time, which precedes  $t_{i-1}$  subject to  $t_{j-1} - \bar{t}_{i-1}^{(-)} > m_1$ . It is noted that  $E_{\Pi}[Z_{1,\ell,ij}^{(\pm)}] = -\gamma_\eta(\ell)$  for all  $\ell$ . A sequence of the  $Z_{1,\ell,ij}^{(\pm)}$  arranged in ascending order of the index  $i$  is defined as  $\{Z_{1,\ell,k}^{(\pm)}\}_{k=1}^{N_{1,\ell}^{(\pm)}}$ . In what follows,  $N_{1,\ell}$  is used instead of  $N_{1,\ell}^{(\pm)}$  for representing the number of observations in the sequence for simplicity. Then under the threshold value  $m_1$  determined through the test statistic (16) and the asymptotic condition that  $M_{1,\ell} \rightarrow \infty$  and  $M_{1,\ell}/N_{1,\ell} \rightarrow 0$  as  $N_{1,\ell}$  goes to infinity, the noise autocovariance estimator and its asymptotic distribution can be described as

$$\hat{\gamma}_\eta(\ell) = -\frac{1}{N_{1,\ell}} \sum_{k=1}^{N_{1,\ell}} Z_{1,\ell,k}^{(\pm)}, \quad \sqrt{N_{1,\ell}}(\hat{\gamma}_\eta(\ell) - \gamma_\eta(\ell)) \xrightarrow{a} N(0, \omega_{\eta,\ell}^2), \quad (19)$$

where  $\omega_{\eta,\ell}^2 = \lim_{N_{1,\ell} \rightarrow \infty} E_{\Pi}[N_{1,\ell}(\hat{\gamma}_\eta(\ell) - \gamma_\eta(\ell))^2]$ . The subsampling variance estimator of  $\omega_{\eta,\ell}^2$  is given by

$$\hat{\omega}_{\eta,\ell}^2 = \frac{M_{1,\ell}}{K_{1,\ell}} \sum_{h=0}^{K_{1,\ell}-1} \left( \bar{Z}_{1,\ell,M_{1,\ell}}^{(\pm),h} - \frac{1}{K_{1,\ell}} \sum_{h=0}^{K_{1,\ell}-1} \bar{Z}_{1,\ell,M_{1,\ell}}^{(\pm),h} \right)^2, \quad (20)$$

where  $\bar{Z}_{1,\ell,M_\ell}^{(\pm),h}$  is the sample mean of the  $h$ th subseries  $\{Z_{1,\ell}^{(\pm),h}\}$ :

$$\{Z_{1,\ell}^{(\pm),h}\} := \left( Z_{1,\ell,hM_{1,\ell}+1}^{(\pm)}, Z_{1,\ell,hM_{1,\ell}+2}^{(\pm)}, \dots, Z_{1,\ell,(h+1)M_{1,\ell}}^{(\pm)} \right), \\ 0 \leq h \leq K_{1,\ell}-1, \quad K_{1,\ell} = [N_{1,\ell}/M_{1,\ell}].$$

It is thus possible to use the following test statistic to determine whether the noise autocovariance is significantly different from zero.

**Corollary 2.** *The test for autocovariance of market microstructure noise.*

- Case of  $\ell = 0$ , test of the noise variance.

Let the null hypothesis and the alternative be  $\gamma_\eta(0) = 0$  and  $\gamma_\eta(0) > 0$ . As  $N_{1,0}$  goes to infinity s.t.  $M_{1,0} \rightarrow \infty$  and  $M_{1,0}/N_{1,0} \rightarrow 0$ , the one-sided test statistic of the noise variance is asymptotically distributed as chi-square with one degree of freedom:

$$\tau_\eta^*(0) = \left( \frac{\sqrt{N_{1,0}} \hat{\gamma}_\eta(0)}{\hat{\omega}_{\eta,0}} \right)^2 \xrightarrow{a} \chi^2(1) \quad (21)$$

under the null hypothesis.  $\tau_\eta^*(0)$  diverges under the alternative.  $\hat{\omega}_{\eta,0}$  is the square root of variance estimator given by (20).

- Case of  $\ell > 0$ .

Let the null hypothesis and the alternative be  $\gamma_\eta(\ell) = 0$  and  $\gamma_\eta(\ell) \neq 0$ . As  $N_{1,\ell}$  goes to infinity s.t.  $M_{1,\ell} \rightarrow \infty$  and  $M_{1,\ell}/N_{1,\ell} \rightarrow 0$ , the test statistic of the noise autocovariance is asymptotically distributed as standard normal:

$$\tau_\eta^*(\ell) := \frac{\sqrt{N_{1,\ell}} \hat{\gamma}_\eta(\ell)}{\hat{\omega}_{\eta,\ell}} \xrightarrow{a} N(0, 1) \quad (22)$$

under the null hypothesis.  $\tau_\eta^*(\ell)$  diverges under the alternative.  $\hat{\omega}_{\eta,\ell}$  is the square root of variance estimator given by (20).

## 4 MONTE CARLO SIMULATION AND EMPIRICAL ILLUSTRATION

This section examines the finite sample properties of the autocovariance and cross-covariance estimators as well as the test statistics through Monte Carlo simulation. The empirical illustration is based on high-frequency transaction prices for a sample of stocks traded on the Osaka Securities Exchange.

### 4.1 Finite Sample Properties of Covariance Estimators of Noises

The data-generating process introduced in Voev and Lunde (2007) is used in the examination of finite sample properties. It is assumed that the equilibrium price processes  $P_1^*$  and  $P_2^*$  follow the stochastic differential equations:

$$dP_y^*(t) = \sigma_y(t) \left[ \sqrt{1 - \lambda_y^2} dW_y^{(A)}(t) + \lambda_y dW_y^{(B)}(t) \right], \quad (23) \\ d\sigma_y^2(t) = \kappa_y(\theta_y - \sigma_y^2(t)) dt + \omega_y \sigma_y^2(t) dW_y^{(B)}(t), \quad y = 1, 2,$$

where  $W_y^{(\cdot)}$  represents a standard Brownian motion and  $\sigma_y^2(t)$  follows the generalized autoregressive conditional heteroskedasticity (GARCH) diffusion process. When  $W_1^{(A)}$  and  $W_2^{(A)}$  are correlated with  $d\langle W_1^{(A)}, W_2^{(A)} \rangle_t = \rho^*(t)dt$ , the anti-Fisher transformation can be used to generate stochastic correlation  $\rho^*(t)$ :

$$\rho^*(t) = \frac{\exp(2x(t)) - 1}{\exp(2x(t)) + 1},$$

$$dx(t) = \kappa_3(\theta_3 - x(t))dt + \omega_3 x(t)dW(t),$$

where  $x(t)$  follows the GARCH diffusion process. The parameter values amount to  $(\lambda_1, \lambda_2) = (0.5, 0.5)$ ,  $(\kappa_1, \kappa_2, \kappa_3) = (0.006, 0.037, 0.051)$ ,  $(\theta_1, \theta_2, \theta_3) = (2.719, 2.527, 0.200)$  and  $(\omega_1, \omega_2, \omega_3) = (0.0382, 0.216, 0.130)$ . The parameters  $\kappa_y$ ,  $\theta_y$ , and  $\omega_y$ , for  $y = 1, 2$  are obtained from GARCH(1,1) estimates with Intel and Microsoft daily returns, using Drost and Werker's (1996) relationship between the continuous time and discrete time parameters. The trading time per day is set as 6.5 hours as with the NYSE and NASDAQ markets.

**4.1.1 Bivariate moving average noise.** This simulation exercise is based on regular and synchronous sampling of bivariate moving average noise processes truncated at some finite lag. The noise  $\eta(t)$  and  $\delta(t)$  terms are generated with the following bivariate MA(2) model for each transaction time:

$$\eta(t) = \epsilon_\eta(t) - 0.5 \epsilon_\eta(t-1) + 0.4 \epsilon_\eta(t-2) - 0.1 \epsilon_\delta(t-1) + 0.15 \epsilon_\delta(t-2),$$

$$\delta(t) = \epsilon_\delta(t) - 0.4 \epsilon_\eta(t-1) + 0.15 \epsilon_\eta(t-2) - 0.1 \epsilon_\delta(t-1) + 0.3 \epsilon_\delta(t-2).$$

The noise-to-signal ratio is defined as the noise variance divided by the integrated variance. Hansen and Lunde (2006) report that noise-to-signal ratios range from 0.0002 to 0.006 using transactions data for 30 stocks traded on the NYSE and NASDAQ markets. The noise variance is set such that the noise-to-signal ratios for assets 1 and 2 amount to 0.005 and 0.002, respectively.

The transaction time intervals are set to 10, 15, and 30 seconds. Two series of transaction prices, which represent the sum of the equilibrium price and noise, are sampled synchronously at equidistant times. In the regular and synchronous sampling,  $\ell = 1$  represents one tick time whose length is equal to the sampling interval (10, 15, or 30 seconds). These sampling intervals are associated with the sample sizes of 2300, 1550, and 770,  $N_\ell$  ( $N_{2,\ell}$ ) for  $\ell > 0$  per day, respectively. On the other hand, the irregular and nonsynchronous sampling is applied in Section 4.1.2.

The test statistics defined in (9) and (16) are used to find the threshold value where noise cross-covariance and autocovariance become zero. It is also possible to compare the performance of the following covariance estimators expressed by (11) and (19).

- The cross and auto covariance estimators  $\hat{\gamma}(\ell)$  and  $\hat{\gamma}_\delta(\ell)$  proposed in (11) and (19). These estimators are unbiased if the threshold values are correctly selected. Since the subsampling variance estimation affects  $\hat{\gamma}(\ell)$  and  $\hat{\gamma}_\delta(\ell)$  through the test statistics (9) and (16) for the threshold value of noise

dependence, we will see the effects using Monte Carlo simulation. For the variance estimation in the test statistic (9) and (16), the asymptotic formula  $M_\ell = cN_\ell^{1/3}$  and  $M_{2,\ell} = cN_{2,\ell}^{1/3}$  for a given value of  $c$  is used to determine the number of observations within each subseries  $M_\ell$  and the total number of subseries  $K_\ell$ . The cross and auto covariance estimators  $\hat{\gamma}(\ell)_{[c]}$  and  $\hat{\gamma}(\ell)_{[c]}$  can be expressed as function of  $c$ , which takes the values of 4, 2, 1, and 0.5 in order to examine its effects. The corresponding values of  $M_\ell$  ( $M_{2,\ell}$ ) and  $K_\ell$  ( $K_{2,\ell}$ ) are:  $(c, M_\ell, K_\ell) = (4, 52, 44)$ ,  $(2, 26, 88)$ ,  $(1, 13, 176)$ ,  $(0.5, 7, 330)$ ;  $(c, M_\ell, K_\ell) = (4, 46, 33)$ ,  $(2, 23, 67)$ ,  $(1, 12, 129)$ ,  $(0.5, 6, 258)$ ; and  $(c, M_\ell, K_\ell) = (4, 36, 21)$ ,  $(2, 18, 42)$ ,  $(1, 9, 85)$ ,  $(0.5, 5, 154)$  for the sample sizes of 2300, 1550, and 770, respectively.

- The cross and auto covariance estimators based on the sample means of  $Z_{\ell,k}$  and  $Z_{2,\ell,k}$ , that is,  $\tilde{\gamma}(\ell) := -\frac{1}{N_\ell} \sum_{k=1}^{N_\ell} Z_{\ell,k}$  and  $\tilde{\gamma}_\delta(\ell) := -\frac{1}{N_{2,\ell}} \sum_{k=1}^{N_{2,\ell}} Z_{2,\ell,k}$ . The conditional expectation of  $Z_{\ell,k}$  is equal to  $\gamma(\ell + \Delta t_i) - \gamma(\ell + \Delta t_i + \Delta s_j) - \gamma(\ell) + \gamma(\ell + \Delta s_j)$  as represented in (5). It is noted that there possibly exist nonzero  $\gamma(\ell + \Delta t_i)$ ,  $\gamma(\ell + \Delta t_i + \Delta s_j)$  and  $\gamma(\ell + \Delta s_j)$  terms. The existence of these nonzero terms is just the difference between the estimators  $\hat{\gamma}(\ell)$  and  $\tilde{\gamma}(\ell)$  because  $\hat{\gamma}(\ell)$  is based on a product returns with intervals extended such that all covariance terms except  $\gamma(\ell)$  become zero. Therefore, the cross-covariance estimator  $\tilde{\gamma}(\ell)$  with  $\ell$  less than the threshold value is biased and the same argument applies for the autocovariance estimator  $\tilde{\gamma}_\delta(\ell)$ .
- The Voev and Lunde's (2007) cross-covariance estimator:

$$\ddot{\gamma}(\ell) = \frac{1}{S_1 S_2} \sum_{b_1=1}^{S_1} \sum_{b_2=1}^{S_2} \ddot{\gamma}(\ell)_{b_1, b_2},$$

where  $S_1$  and  $S_2$  represent the number of subgrids for asset 1 and 2, respectively. This cross-covariance estimator  $\ddot{\gamma}(\ell)$  is based on subsampling method. The approach proposed by Voev and Lunde (2007) does not include the selection of the threshold value prior to the estimation of the cross-covariance. Instead, the length of subsampled intervals is set to be large enough to eliminate any bias in the covariances  $\gamma(\ell + \Delta t_i)$ ,  $\gamma(\ell + \Delta t_i + \Delta s_j)$  and  $\gamma(\Delta s_j)$ . For the  $b_y$ th subgrid ( $b_y = 1, \dots, S_y$ , for  $y = 1, 2$ ), it is possible to define the subgrid returns as  $r_{1,i^b} := P_1(t_i^b) - P_1(t_{i^b-1})$  with  $\{t_i^b\}_{i^b=1}^{N_{1,\ell}^b}$  and  $r_{2,j^b} := P_2(s_{j^b}) - P_2(s_{j^b-1})$  with  $\{s_{j^b}\}_{j^b=1}^{N_{2,\ell}^b}$ , where  $b$  is used in place of  $b_1$  and  $b_2$  for expository reasons. Then  $\ddot{\gamma}(\ell)_b$  can be expressed as:

$$\ddot{\gamma}(\ell)_b = \begin{cases} -\frac{1}{N_{\ell,b}} \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} r_{1,i^b} r_{2,j^b} 1_{\{t_{i^b-1} - s_{j^b} = \ell\}} & \text{for } \ell > 0 \\ -\frac{1}{N_{\ell,b}} \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} r_{1,i^b} r_{2,j^b} 1_{\{t_{i^b-1} - s_{j^b} = 0\} \cup \{s_{j^b-1} - t_{i^b} = 0\}} & \text{for } \ell = 0 \quad (24) \\ -\frac{1}{N_{\ell,b}} \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} r_{1,i^b} r_{2,j^b} 1_{\{s_{j^b-1} - t_{i^b} = -\ell\}} & \text{for } \ell < 0 \end{cases}$$



with

$$N_{\ell,b} = \begin{cases} \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} 1_{\{t_{i^b-1}-s_{j^b}=\ell\}} & \text{for } \ell > 0 \\ \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} 1_{\{t_{i^b-1}-s_{j^b}=0\} \cup \{s_{j^b-1}-t_{i^b}=0\}} & \text{for } \ell = 0 \\ \sum_{i^b=1}^{N_{1,\ell}^b} \sum_{j^b=1}^{N_{2,\ell}^b} 1_{\{s_{j^b-1}-t_{i^b}=-\ell\}} & \text{for } \ell < 0. \end{cases} \quad (25)$$

On the other hand, the covariance estimators proposed in (11) and (19) are based on the interval with appropriate length where the dependence of the noise disappears. The length of the interval is determined through the test statistic (9) and (16).

Following Voev and Lunde (2007), the significance of noise cross-covariance can be assessed using the test statistic  $\tau^*(\ell)_{VL} = \check{\gamma}(\ell)/\sqrt{V[\check{\gamma}(\ell)]}$ . This test statistic uses an approximated variance of cross-covariance estimator under the assumption of i.i.d. noise. The approximated variance can be expressed as:

$$V[\check{\gamma}(\ell)] \approx \begin{cases} \frac{1}{(S_1 S_2)^2} \sum_{b=1}^{S_1 S_2} \frac{1}{N_{\ell,b}} (v_x + 5\sigma_\eta^2 \sigma_\delta^2) & \text{for } \ell = 0 \\ \frac{1}{(S_1 S_2)^2} \sum_{b=1}^{S_1 S_2} \frac{1}{N_{\ell,b}} (v_x + 4\sigma_\eta^2 \sigma_\delta^2) & \text{for } \ell \neq 0, \end{cases} \quad (26)$$

where  $v_x := \sigma_{x \text{ sec},1}^2 \sigma_{x \text{ sec},2}^2 + 2\sigma_\delta^2 \sigma_{x \text{ sec},1}^2 + 2\sigma_\eta^2 \sigma_{x \text{ sec},2}^2$ ,  $\sigma_{x \text{ sec},y}^2 = \frac{\int_0^1 \sigma_y^2(s) ds}{23400/x}$  for  $y = 1, 2$  is the  $x$ -second average integrated variance of each asset 1 and 2. The noise variances  $\sigma_\eta^2$  and  $\sigma_\delta^2$  are estimated by subsampling the realized variance with one-minute returns and subsequently dividing by twice the number of returns. The integrated variance is calculated using the realized kernel-based estimator of Barndorff-Nielsen et al. (2008). On the other hand, the test statistics derived in this paper are constructed by the subsampling variance estimator, which does not require the i.i.d. noise approximation.

In contrast to the approach suggested by Voev and Lunde (2007), the cross and auto covariance estimators (11) and (19) proposed in the present study require the prior knowledge of the threshold values of noise dependence. The adequacy of these threshold values is assessed using the test statistic (9) and (16). For the bivariate MA(2) noise process, the true threshold values  $m^+$ ,  $m^-$ , and  $m_2$  are found to be 2 tick times ( $|\ell| = 2$ ). This result implies that as long as the selected threshold value is not less than 2 ticks, the proposed covariance estimators  $\hat{\gamma}(\ell)$  and  $\hat{\gamma}_\delta(\ell)$  are unbiased. Based on 1000 iterations, Table 1 represents the probability of threshold-selection errors in the case where the estimated threshold values  $\hat{m}^+$ ,  $\hat{m}^-$ , and  $\hat{m}_2$  are less than 2 ticks. Although the probability of selection errors with  $c = 0.5$  is lowest across the sampling intervals, the difference in probability relative to other  $c$  values is rather small. Furthermore, the probability of selection errors approaches zero as the time interval becomes shorter, and the sample size increases.

The effects of threshold selection errors are reported in Tables 2 and 3. With a sample size of 770 corresponding to the sampling frequency of 30 seconds, there is a bias in the estimation of  $\hat{\gamma}(\ell)_{[c]}$  and  $\hat{\gamma}_\delta(\ell)_{[c]}$  for any given value of  $c$ . However,

**Table 1** Probability of the estimated threshold value being less than 2

|                         | 5 sec | 10 sec | 15 sec | 30 sec |
|-------------------------|-------|--------|--------|--------|
| Prob( $\hat{m}^+ < 2$ ) |       |        |        |        |
| $c = 4$                 | 0.013 | 0.166  | 0.299  | 0.438  |
| $c = 2$                 | 0.016 | 0.178  | 0.305  | 0.469  |
| $c = 1$                 | 0.013 | 0.171  | 0.294  | 0.469  |
| $c = 0.5$               | 0.011 | 0.141  | 0.271  | 0.421  |
| Prob( $\hat{m}^- < 2$ ) |       |        |        |        |
| $c = 4$                 | 0.005 | 0.108  | 0.222  | 0.424  |
| $c = 2$                 | 0.006 | 0.114  | 0.248  | 0.426  |
| $c = 1$                 | 0.003 | 0.098  | 0.239  | 0.424  |
| $c = 0.5$               | 0.001 | 0.074  | 0.193  | 0.394  |
| Prob( $\hat{m}_2 < 2$ ) |       |        |        |        |
| $c = 4$                 | 0.007 | 0.127  | 0.243  | 0.406  |
| $c = 2$                 | 0.004 | 0.124  | 0.262  | 0.447  |
| $c = 1$                 | 0.002 | 0.122  | 0.238  | 0.442  |
| $c = 0.5$               | 0.001 | 0.111  | 0.209  | 0.402  |

This table represents the probability of threshold-selection errors in the case where the estimated threshold values  $\hat{m}^+$ ,  $\hat{m}^-$ , and  $\hat{m}_2$  are less than 2 ticks. This is for the sake of the examination about the adequacy of threshold values assessed using the test statistics (9) and (16). For the bivariate MA(2) noise process, the true threshold values  $m^+$ ,  $m^-$ , and  $m_2$  are found to be 2 tick times ( $|\ell| = 2$ ). For the variance estimation in the test statistics (9) and (16), the asymptotic formula  $M_\ell = cN_\ell^{1/3}$  and  $M_{2,\ell} = cN_{2,\ell}^{1/3}$  for a given value of  $c$  is used to determine the number of observations within each subseries  $M_\ell$  and the total number of subseries  $K_\ell$ . The transaction time intervals are set to 5, 10, 15, and 30 seconds.

this bias approaches zero with respect to higher trading intensities such as 10 seconds since the probability of selection errors decreases substantially. Sampling over a period of few days is required in the case of lower trading intensity in order to obtain a larger sample size. The fourth panel in Table 2 suggests that the magnitude of bias in  $\hat{\gamma}(\ell)_{[c]}$  and  $\hat{\gamma}_8(\ell)_{[c]}$  estimators using a 3-day period with the sampling frequency of 30 seconds is smaller than that of periods of just one day, as reported in the third panel. Judging from the reported results, the cross and auto covariance estimators proposed in this study are more reliable when the sampling is made over a period of few days even with lower trading intensities.

It is clear that the estimators  $\tilde{\gamma}(\ell)$  and  $\tilde{\gamma}_8(\ell)$  with  $\ell = 0, 1$  are biased as mentioned above. The bias of Voev and Lunde's (2007) cross-covariance estimator  $\tilde{\gamma}(\ell)$  is virtually zero in all cases because the subsampling returns are observed with intervals that are large enough to preclude selection errors. However, it is noted that the RMSE of  $\tilde{\gamma}(\ell)$  is larger than  $\hat{\gamma}(\ell)_{[c]}$  for any given value of  $c$  and that the RMSE ratio of  $\tilde{\gamma}(\ell)$  to  $\hat{\gamma}(\ell)_{[1]}$  is larger than 1, as indicated in Table 3. The higher

**Table 2** Bias of cross and auto covariance estimators of bivariate MA noise processes

| True covariances                                 |                |                         |                        |                                 | $c = 4$                    |                                     | $c = 2$                    |                                     | $c = 1$                    |                                     | $c = 0.5$                    |                                       | $\check{\gamma}(\ell)$ |
|--------------------------------------------------|----------------|-------------------------|------------------------|---------------------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|------------------------------|---------------------------------------|------------------------|
| $\ell$                                           | $\gamma(\ell)$ | $\gamma_{\delta}(\ell)$ | $\check{\gamma}(\ell)$ | $\check{\gamma}_{\delta}(\ell)$ | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}_{\delta}(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_{\delta}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_{\delta}(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_{\delta}(\ell)_{[0.5]}$ | $\check{\gamma}(\ell)$ |
| Sample size: 2300; sampling interval: 10 seconds |                |                         |                        |                                 |                            |                                     |                            |                                     |                            |                                     |                              |                                       |                        |
| -4                                               | 0              |                         | -0.0001                |                                 | -0.0002                    | -0.0002                             | -0.0002                    | -0.0002                             | -0.0002                    | -0.0002                             | -0.0001                      | -0.0001                               | -0.0008                |
| -3                                               | 0              |                         | 0.0001                 |                                 | 0.0003                     | 0.0003                              | 0.0003                     | 0.0003                              | 0.0003                     | 0.0003                              | 0.0002                       | 0.0002                                | -0.0008                |
| -2                                               | 0.0088         |                         | 0.0000                 |                                 | -0.0006                    | -0.0006                             | -0.0006                    | -0.0006                             | -0.0006                    | -0.0006                             | -0.0005                      | -0.0005                               | -0.0007                |
| -1                                               | -0.0180        |                         | -0.0177                |                                 | -0.0012                    | -0.0012                             | -0.0013                    | -0.0013                             | -0.0010                    | -0.0010                             | -0.0007                      | -0.0007                               | -0.0007                |
| 0                                                | 0.0269         | 0.0217                  | 0.0415                 | 0.0290                          | -0.0019                    | -0.0014                             | -0.0020                    | -0.0013                             | -0.0018                    | -0.0013                             | -0.0014                      | -0.0011                               | -0.0004                |
| 1                                                | -0.0151        | -0.0115                 | -0.0160                | -0.0121                         | -0.0018                    | -0.0011                             | -0.0020                    | -0.0010                             | -0.0019                    | -0.0010                             | -0.0016                      | -0.0009                               | 0.0000                 |
| 2                                                | 0.0080         | 0.0061                  | 0.0001                 | 0.0000                          | -0.0007                    | -0.0004                             | -0.0008                    | -0.0004                             | -0.0007                    | -0.0004                             | -0.0006                      | -0.0004                               | 0.0001                 |
| 3                                                | 0              | 0                       | -0.0001                | 0.0000                          | 0.0003                     | 0.0002                              | 0.0004                     | 0.0002                              | 0.0003                     | 0.0002                              | 0.0003                       | 0.0002                                | 0.0000                 |
| 4                                                | 0              | 0                       | 0.0000                 | 0.0000                          | -0.0001                    | -0.0001                             | -0.0002                    | -0.0001                             | -0.0002                    | -0.0001                             | -0.0001                      | -0.0001                               | 0.0001                 |
| Sample size: 1550; sampling interval: 15 seconds |                |                         |                        |                                 |                            |                                     |                            |                                     |                            |                                     |                              |                                       |                        |
| -4                                               | 0              |                         | -0.0001                |                                 | -0.0003                    | -0.0003                             | -0.0003                    | -0.0003                             | -0.0003                    | -0.0003                             | -0.0003                      | -0.0003                               | 0.0001                 |
| -3                                               | 0              |                         | 0.0001                 |                                 | 0.0006                     | 0.0006                              | 0.0007                     | 0.0007                              | 0.0007                     | 0.0007                              | 0.0006                       | 0.0006                                | -0.0001                |
| -2                                               | 0.0088         |                         | 0.0000                 |                                 | -0.0011                    | -0.0011                             | -0.0012                    | -0.0012                             | -0.0012                    | -0.0012                             | -0.0010                      | -0.0010                               | 0.0001                 |
| -1                                               | -0.0180        |                         | -0.0177                |                                 | -0.0028                    | -0.0028                             | -0.0032                    | -0.0032                             | -0.0031                    | -0.0031                             | -0.0024                      | -0.0024                               | 0.0000                 |
| 0                                                | 0.0269         | 0.0217                  | 0.0415                 | 0.0291                          | -0.0037                    | -0.0028                             | -0.0040                    | -0.0029                             | -0.0037                    | -0.0025                             | -0.0032                      | -0.0022                               | 0.0000                 |
| 1                                                | -0.0151        | -0.0115                 | -0.0159                | -0.0121                         | -0.0034                    | -0.0024                             | -0.0037                    | -0.0025                             | -0.0034                    | -0.0022                             | -0.0032                      | -0.0019                               | -0.0001                |
| 2                                                | 0.0080         | 0.0061                  | -0.0001                | -0.0001                         | -0.0012                    | -0.0009                             | -0.0012                    | -0.0010                             | -0.0013                    | -0.0009                             | -0.0012                      | -0.0009                               | -0.0002                |
| 3                                                | 0              | 0                       | 0.0001                 | 0.0001                          | 0.0009                     | 0.0002                              | 0.0008                     | 0.0003                              | 0.0009                     | 0.0004                              | 0.0008                       | 0.0004                                | -0.0002                |
| 4                                                | 0              | 0                       | 0.0000                 | -0.0001                         | -0.0004                    | -0.0004                             | -0.0003                    | -0.0003                             | -0.0003                    | -0.0003                             | -0.0004                      | -0.0003                               | -0.0002                |

(continued)

(continued)

Table 2 (continued)

| True covariances                                           |                | $c = 4$              |                        |                            | $c = 2$                    |                            |                            | $c = 1$                    |                            |                              | $c = 0.5$                    |                      |                      |
|------------------------------------------------------------|----------------|----------------------|------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|----------------------------|------------------------------|------------------------------|----------------------|----------------------|
| $\ell$                                                     | $\gamma(\ell)$ | $\hat{\gamma}(\ell)$ | $\tilde{\gamma}(\ell)$ | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\dot{\gamma}(\ell)$ | $\dot{\gamma}(\ell)$ |
| Sample size: 770; sampling interval: 30 seconds            |                |                      |                        |                            |                            |                            |                            |                            |                            |                              |                              |                      |                      |
| -4                                                         | 0              | -0.0001              |                        | -0.0002                    | -0.0003                    | -0.0002                    | -0.0002                    |                            |                            | -0.0003                      |                              | -0.0004              |                      |
| -3                                                         | 0              | 0.0000               |                        | 0.0010                     | 0.0010                     | 0.0009                     | 0.0009                     |                            |                            | 0.0009                       |                              | -0.0003              |                      |
| -2                                                         | 0.0088         | 0.0001               |                        | -0.0011                    | -0.0012                    | -0.0011                    | -0.0011                    |                            |                            | -0.0012                      |                              | -0.0005              |                      |
| -1                                                         | -0.0180        | -0.0179              |                        | -0.0059                    | -0.0059                    | -0.0058                    | -0.0058                    |                            |                            | -0.0055                      |                              | -0.0007              |                      |
| 0                                                          | 0.0269         | 0.0217               | 0.0290                 | -0.0062                    | -0.0064                    | -0.0051                    | -0.0063                    | -0.0048                    |                            | -0.0058                      | -0.0043                      | -0.0005              |                      |
| 1                                                          | -0.0151        | -0.0115              | -0.0160                | -0.0053                    | -0.0040                    | -0.0058                    | -0.0046                    | -0.0059                    | -0.0044                    | -0.0052                      | -0.0039                      | -0.0005              |                      |
| 2                                                          | 0.0080         | 0.0061               | 0.0000                 | -0.0014                    | -0.0011                    | -0.0015                    | -0.0013                    | -0.0015                    | -0.0010                    | -0.0014                      | -0.0009                      | -0.0006              |                      |
| 3                                                          | 0              | 0.0002               | 0.0001                 | 0.0007                     | 0.0004                     | 0.0009                     | 0.0003                     | 0.0009                     | 0.0005                     | 0.0008                       | 0.0005                       | -0.0004              |                      |
| 4                                                          | 0              | -0.0002              | -0.0001                | -0.0002                    | -0.0004                    | -0.0003                    | -0.0005                    | -0.0004                    | -0.0003                    | -0.0004                      | -0.0002                      | -0.0007              |                      |
| Sample size: 2300; sampling interval: 30 seconds in 3 days |                |                      |                        |                            |                            |                            |                            |                            |                            |                              |                              |                      |                      |
| -4                                                         | 0              | -0.0001              |                        | -0.0002                    | -0.0002                    | -0.0002                    | -0.0002                    |                            |                            | -0.0002                      |                              | 0.0001               |                      |
| -3                                                         | 0              | 0.0001               |                        | 0.0003                     | 0.0004                     | 0.0004                     | 0.0004                     |                            |                            | 0.0003                       |                              | 0.0002               |                      |
| -2                                                         | 0.0088         | 0.0000               |                        | -0.0007                    | -0.0008                    | -0.0007                    | -0.0007                    |                            |                            | -0.0007                      |                              | 0.0002               |                      |
| -1                                                         | -0.0180        | -0.0177              |                        | -0.0014                    | -0.0015                    | -0.0014                    | -0.0014                    |                            |                            | -0.0010                      |                              | 0.0002               |                      |
| 0                                                          | 0.0269         | 0.0217               | 0.0290                 | -0.0022                    | -0.0015                    | -0.0022                    | -0.0015                    | -0.0021                    | -0.0016                    | -0.0019                      | -0.0016                      | -0.0001              |                      |
| 1                                                          | -0.0151        | -0.0115              | -0.0160                | -0.0022                    | -0.0012                    | -0.0021                    | -0.0012                    | -0.0021                    | -0.0013                    | -0.0020                      | -0.0013                      | -0.0004              |                      |
| 2                                                          | 0.0080         | 0.0061               | 0.0000                 | -0.0008                    | -0.0005                    | -0.0009                    | -0.0005                    | -0.0009                    | -0.0005                    | -0.0008                      | -0.0005                      | -0.0004              |                      |
| 3                                                          | 0              | -0.0001              | 0.0000                 | 0.0003                     | 0.0003                     | 0.0004                     | 0.0002                     | 0.0004                     | 0.0002                     | 0.0002                       | 0.0001                       | -0.0004              |                      |
| 4                                                          | 0              | 0.0000               | 0.0000                 | -0.0002                    | -0.0001                    | -0.0002                    | -0.0001                    | -0.0003                    | -0.0001                    | -0.0002                      | -0.0002                      | -0.0004              |                      |

The market microstructure noises are generated by a bivariate MA(2) process.  $\ell = 1$  represents one tick whose length is equal to each sampling interval (10, 15, and 30 seconds). The second and third columns represent true cross and auto covariances  $\gamma(\ell)$ ,  $\tilde{\gamma}(\ell)$ .

**Table 3** RMSE of cross and auto covariance estimators of bivariate MA noise processes

| True covariances                                 |                |                  | $c = 4$              |                              | $c = 2$                    |                              | $c = 1$                    |                              | $c = 0.5$                    |                                | RMSE ratio           |                                                         |
|--------------------------------------------------|----------------|------------------|----------------------|------------------------------|----------------------------|------------------------------|----------------------------|------------------------------|------------------------------|--------------------------------|----------------------|---------------------------------------------------------|
| $\ell$                                           | $\gamma(\ell)$ | $\gamma_s(\ell)$ | $\hat{\gamma}(\ell)$ | $\hat{\gamma}_s(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_s(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_s(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_s(\ell)_{[0.5]}$ | $\hat{\gamma}(\ell)$ | $\frac{\hat{\gamma}(\ell)}{\hat{\gamma}(\ell)_{[0.5]}}$ |
| Sample size: 2300; sampling interval: 10 seconds |                |                  |                      |                              |                            |                              |                            |                              |                              |                                |                      |                                                         |
| -4                                               | 0              |                  | 0.0031               |                              | 0.0020                     | 0.0021                       | 0.0020                     |                              | 0.0020                       |                                | 0.0156               | 7.6173                                                  |
| -3                                               | 0              |                  | 0.0032               |                              | 0.0023                     | 0.0023                       | 0.0023                     |                              | 0.0022                       |                                | 0.0158               | 6.9754                                                  |
| -2                                               | 0.0088         |                  | 0.0032               | 0.0025                       | 0.0026                     | 0.0026                       | 0.0025                     |                              | 0.0025                       |                                | 0.0156               | 6.1666                                                  |
| -1                                               | -0.0180        |                  | 0.0180               | 0.0054                       | 0.0054                     | 0.0054                       | 0.0051                     |                              | 0.0046                       |                                | 0.0158               | 3.0850                                                  |
| 0                                                | 0.0269         | 0.0217           | 0.0416               | 0.0057                       | 0.0059                     | 0.0048                       | 0.0056                     | 0.0048                       | 0.0052                       | 0.0046                         | 0.0121               | 2.1403                                                  |
| 1                                                | -0.0151        | -0.0115          | 0.0163               | 0.0059                       | 0.0041                     | 0.0061                       | 0.0060                     | 0.0040                       | 0.0055                       | 0.0039                         | 0.0152               | 2.5456                                                  |
| 2                                                | 0.0080         | 0.0061           | 0.0032               | 0.0025                       | 0.0021                     | 0.0025                       | 0.0025                     | 0.0020                       | 0.0024                       | 0.0021                         | 0.0152               | 6.1137                                                  |
| 3                                                | 0              | 0                | 0.0032               | 0.0024                       | 0.0020                     | 0.0024                       | 0.0019                     | 0.0024                       | 0.0019                       | 0.0023                         | 0.0152               | 6.4281                                                  |
| 4                                                | 0              | 0                | 0.0032               | 0.0021                       | 0.0019                     | 0.0021                       | 0.0018                     | 0.0021                       | 0.0020                       | 0.0018                         | 0.0153               | 7.4416                                                  |
| Sample size: 1550; sampling interval: 15 seconds |                |                  |                      |                              |                            |                              |                            |                              |                              |                                |                      |                                                         |
| -4                                               | 0              |                  | 0.0039               | 0.0031                       | 0.0031                     | 0.0031                       | 0.0031                     |                              | 0.0031                       |                                | 0.0165               | 5.3057                                                  |
| -3                                               | 0              |                  | 0.0039               | 0.0034                       | 0.0033                     | 0.0033                       | 0.0033                     |                              | 0.0034                       |                                | 0.0162               | 4.8833                                                  |
| -2                                               | 0.0088         |                  | 0.0039               | 0.0035                       | 0.0037                     | 0.0037                       | 0.0036                     |                              | 0.0036                       |                                | 0.0164               | 4.5473                                                  |
| -1                                               | -0.0180        |                  | 0.0182               | 0.0080                       | 0.0082                     | 0.0082                       | 0.0080                     |                              | 0.0074                       |                                | 0.0162               | 2.0107                                                  |
| 0                                                | 0.0269         | 0.0217           | 0.0418               | 0.0081                       | 0.0083                     | 0.0071                       | 0.0082                     | 0.0069                       | 0.0077                       | 0.0068                         | 0.0126               | 1.5406                                                  |
| 1                                                | -0.0151        | -0.0115          | 0.0164               | 0.0082                       | 0.0083                     | 0.0062                       | 0.0082                     | 0.0060                       | 0.0079                       | 0.0058                         | 0.0162               | 1.9887                                                  |
| 2                                                | 0.0080         | 0.0061           | 0.0040               | 0.0036                       | 0.0032                     | 0.0036                       | 0.0037                     | 0.0032                       | 0.0037                       | 0.0032                         | 0.0159               | 4.3586                                                  |
| 3                                                | 0              | 0                | 0.0040               | 0.0035                       | 0.0033                     | 0.0034                       | 0.0031                     | 0.0034                       | 0.0035                       | 0.0032                         | 0.0161               | 4.7868                                                  |
| 4                                                | 0              | 0                | 0.0039               | 0.0033                       | 0.0032                     | 0.0030                       | 0.0033                     | 0.0030                       | 0.0034                       | 0.0030                         | 0.0160               | 4.8777                                                  |

(continued)

Table 3 (continued)

| True covariances                                           |                | c = 4                 |                      |                             | c = 2                      |                                   |                            | c = 1                             |                            |                                   | c = 0.5                      |                                     |                      | RMSE ratio                                            |
|------------------------------------------------------------|----------------|-----------------------|----------------------|-----------------------------|----------------------------|-----------------------------------|----------------------------|-----------------------------------|----------------------------|-----------------------------------|------------------------------|-------------------------------------|----------------------|-------------------------------------------------------|
| $\ell$                                                     | $\gamma(\ell)$ | $\gamma_\delta(\ell)$ | $\hat{\gamma}(\ell)$ | $\hat{\gamma}_\delta(\ell)$ | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}_\delta(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_\delta(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_\delta(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_\delta(\ell)_{[0.5]}$ | $\hat{\gamma}(\ell)$ | $\frac{\hat{\gamma}(\ell)}{\hat{\gamma}(\ell)_{[1]}}$ |
| Sample size: 770; sampling interval: 30 seconds            |                |                       |                      |                             |                            |                                   |                            |                                   |                            |                                   |                              |                                     |                      |                                                       |
| -4                                                         | 0              |                       | 0.0057               |                             | 0.0071                     |                                   | 0.0071                     |                                   | 0.0071                     |                                   | 0.0073                       |                                     | 0.0165               | 2.3181                                                |
| -3                                                         | 0              |                       | 0.0057               |                             | 0.0073                     |                                   | 0.0072                     |                                   | 0.0073                     |                                   | 0.0074                       |                                     | 0.0163               | 2.2304                                                |
| -2                                                         | 0.0088         |                       | 0.0057               |                             | 0.0072                     |                                   | 0.0072                     |                                   | 0.0073                     |                                   | 0.0075                       |                                     | 0.0165               | 2.2569                                                |
| -1                                                         | -0.0180        |                       | 0.0189               |                             | 0.0129                     |                                   | 0.0128                     |                                   | 0.0128                     |                                   | 0.0124                       |                                     | 0.0162               | 1.2608                                                |
| 0                                                          | 0.0269         | 0.0217                | 0.0422               | 0.0295                      | 0.0119                     | 0.0128                            | 0.0120                     | 0.0126                            | 0.0120                     | 0.0127                            | 0.0118                       | 0.0125                              | 0.0128               | 1.0720                                                |
| 1                                                          | -0.0151        | -0.0115               | 0.0169               | 0.0128                      | 0.0127                     | 0.0115                            | 0.0125                     | 0.0112                            | 0.0125                     | 0.0113                            | 0.0123                       | 0.0111                              | 0.0167               | 1.3340                                                |
| 2                                                          | 0.0080         | 0.0061                | 0.0056               | 0.0041                      | 0.0072                     | 0.0082                            | 0.0069                     | 0.0077                            | 0.0066                     | 0.0078                            | 0.0070                       | 0.0079                              | 0.0164               | 2.4911                                                |
| 3                                                          | 0              | 0                     | 0.0057               | 0.0041                      | 0.0077                     | 0.0085                            | 0.0072                     | 0.0080                            | 0.0071                     | 0.0081                            | 0.0075                       | 0.0082                              | 0.0168               | 2.3509                                                |
| 4                                                          | 0              | 0                     | 0.0058               | 0.0041                      | 0.0071                     | 0.0081                            | 0.0068                     | 0.0078                            | 0.0067                     | 0.0078                            | 0.0070                       | 0.0078                              | 0.0162               | 2.4104                                                |
| Sample size: 2300; sampling interval: 30 seconds in 3 days |                |                       |                      |                             |                            |                                   |                            |                                   |                            |                                   |                              |                                     |                      |                                                       |
| -4                                                         | 0              |                       | 0.0033               |                             | 0.0039                     |                                   | 0.0038                     |                                   | 0.0039                     |                                   | 0.0039                       |                                     | 0.0095               | 2.4521                                                |
| -3                                                         | 0              |                       | 0.0033               |                             | 0.0040                     |                                   | 0.0041                     |                                   | 0.0041                     |                                   | 0.0042                       |                                     | 0.0095               | 2.3299                                                |
| -2                                                         | 0.0088         |                       | 0.0033               |                             | 0.0040                     |                                   | 0.0040                     |                                   | 0.0040                     |                                   | 0.0041                       |                                     | 0.0095               | 2.3448                                                |
| -1                                                         | -0.0180        |                       | 0.0180               |                             | 0.0068                     |                                   | 0.0068                     |                                   | 0.0067                     |                                   | 0.0064                       |                                     | 0.0095               | 1.4298                                                |
| 0                                                          | 0.0269         | 0.0217                | 0.0417               | 0.0292                      | 0.0067                     | 0.0067                            | 0.0068                     | 0.0065                            | 0.0067                     | 0.0066                            | 0.0064                       | 0.0067                              | 0.0072               | 1.0714                                                |
| 1                                                          | -0.0151        | -0.0115               | 0.0163               | 0.0124                      | 0.0072                     | 0.0057                            | 0.0070                     | 0.0056                            | 0.0071                     | 0.0056                            | 0.0069                       | 0.0057                              | 0.0093               | 1.3175                                                |
| 2                                                          | 0.0080         | 0.0061                | 0.0033               | 0.0024                      | 0.0039                     | 0.0041                            | 0.0037                     | 0.0038                            | 0.0036                     | 0.0039                            | 0.0037                       | 0.0040                              | 0.0091               | 2.4818                                                |
| 3                                                          | 0              | 0                     | 0.0033               | 0.0023                      | 0.0040                     | 0.0042                            | 0.0038                     | 0.0040                            | 0.0038                     | 0.0040                            | 0.0039                       | 0.0042                              | 0.0091               | 2.3827                                                |
| 4                                                          | 0              | 0                     | 0.0033               | 0.0024                      | 0.0038                     | 0.0042                            | 0.0035                     | 0.0039                            | 0.0035                     | 0.0039                            | 0.0036                       | 0.0041                              | 0.0092               | 2.6264                                                |

The market microstructure noises are generated by a bivariate MA(2) processes.  $\ell = 1$  represents one tick time whose length is equal to each sampling interval (10, 15, and 30 seconds). The second and third columns represent true cross and auto covariances  $\gamma(\ell)$ ,  $\gamma_\delta(\ell)$ .

RMSE value of  $\check{\gamma}(\ell)$  is reflective of the larger standard deviation induced by longer intervals used for the estimation of  $\check{\gamma}(\ell)$  relative to  $\hat{\gamma}(\ell)_{[1]}$ .

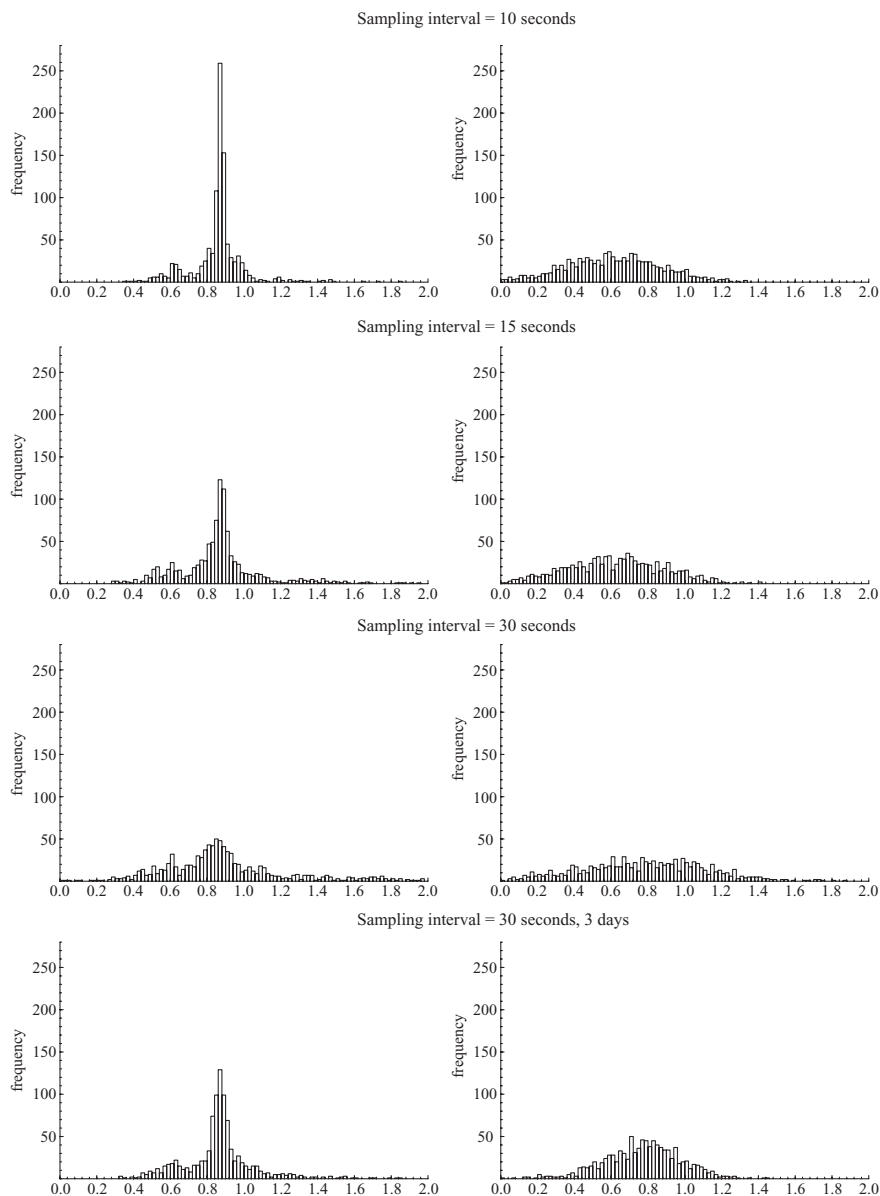
The larger standard deviation of covariance estimator affects also the behavior of the noise cross and auto correlations. It is possible to compare the cross-correlations computed on the basis of the proposed variance and covariance estimators  $\hat{\rho}(\ell) = \hat{\gamma}(\ell)_{[1]} / \sqrt{\hat{\gamma}_\eta(0)_{[1]} \hat{\gamma}_\delta(0)_{[1]}}$ , against those implied by Voev and Lunde (2007). The cross-correlations following Voev and Lunde (2007) are computed as  $\check{\rho}(\ell) = \check{\gamma}(\ell) / (\hat{\sigma}_\eta \hat{\sigma}_\delta)$  where  $\hat{\sigma}_\eta^2$  and  $\hat{\sigma}_\delta^2$  are estimates of the noise variances. Again, the noise variances are based on the subsampling of realized variance with one-minute returns, divided by twice the number of returns. Figures 4 and 5 describe the empirical distributions of cross-correlations in the cases of true cross-correlations amounting to  $\rho(0) = 0.87$  and  $\rho(2) = 0.26$ . The evidence suggests that with respect to all sampling intervals, the cross-correlation estimates  $\hat{\rho}(\ell)$  have a higher degree of concentration around the true correlation value than  $\check{\rho}(\ell)$  based on the estimation approach by Voev and Lunde (2007).

Also, Table 4 reports the size and power of test statistics for significance of the cross and auto covariance estimators proposed in (14), (21), (22) relative to the corresponding estimates following Voev and Lunde (2007). The size of the test statistic proposed in this study approaches the standard values, but that of Voev and Lunde's (2007) test statistic  $\tau^*(\ell)_{VL}$  is rather large. Figure 6 describes the asymptotic distribution (standard normal) and the empirical distributions of  $\hat{\gamma}(\ell)_{[1]}$  and  $\check{\gamma}(\ell)$ , which are standardized by  $\hat{\omega}_\ell$  and  $\sqrt{\hat{V}[\check{\gamma}(\ell)]}$ . Although the standardized empirical distribution of  $\hat{\gamma}(\ell)_{[1]}$  is close to the standard normal distribution, that of  $\check{\gamma}(\ell)$  is heavily tailed. The power of the proposed test statistic increases also as the sample size increases.

The focus is also made on the effects of  $M_\ell$  and  $K_\ell$  selection at fixed  $N_\ell$  under  $M_\ell = cN_\ell^{1/3}$  on the variance estimation. Although the value of  $c$  should be determined in accordance with the degree of noise dependence, the latter is not known a priori. It is also difficult to determine the optimal value of  $c$  in this simulation setting. As a general rule, however,  $c$  takes larger values when the noise process is characterized by stronger dependence. It is conceivable that the behavior of covariance estimators and test statistics is also influenced by the value of  $c$ , but the evidence from Tables 2, 3, and 4 indicates that changes in this parameter do not alter the proposed statistics significantly.

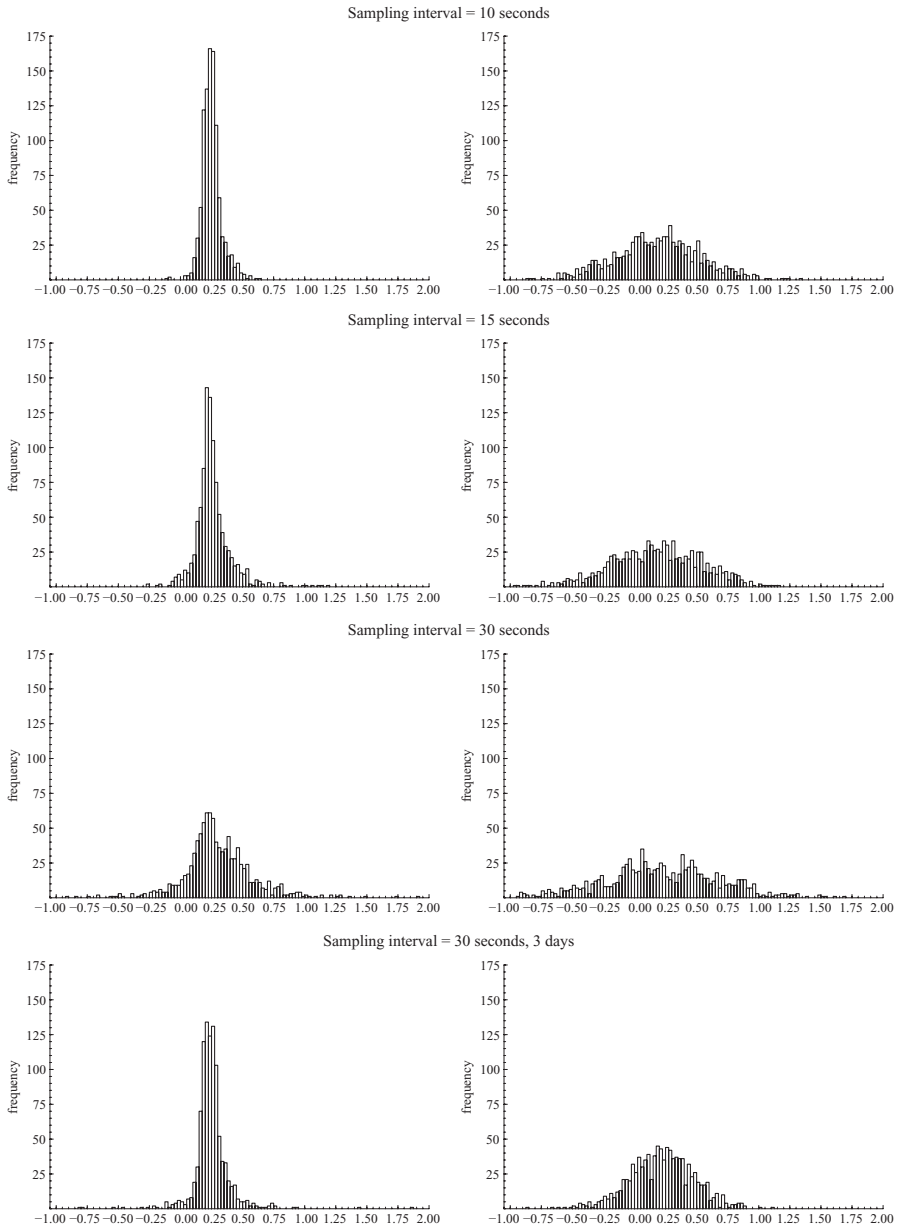
It was assumed thus far that equilibrium prices follow an Itô process without drift as described in (1). It is also important to consider the case where equilibrium prices follow an Itô process with drift. It is noted that the product of the noise terms dominates that of drift terms as the interval length approaches zero. However, it is not clear whether the influence of the drift is negligible because the estimators and test statistics proposed in this paper are based on the expanded intervals in (10) and (18). Thus, the simulation tests are performed again with a drift term added to the stochastic differential equations:

$$dP_y^*(t) = \mu_y dt + \sigma_y(t) \left[ \sqrt{1 - \lambda_y^2} dW_y^{(A)}(t) + \lambda_y dW_y^{(B)}(t) \right], \quad y = 1, 2, \quad (27)$$



**Figure 4** Empirical distributions of  $\hat{\rho}(0)$  (left side) and  $\hat{\rho}(0)$  (right side), where true cross-correlation  $\rho(0) = 0.87$ .





**Figure 5** Empirical distributions of  $\hat{\rho}(2)$  (left side) and  $\check{\rho}(2)$  (right side), where true cross-correlation  $\rho(2) = 0.26$ .

Table 4 Size and power of test statistics for significance in (14), (21), and (22)

| True covariances and correlations                |                |              |                         |                       |                            |                                     |                            |                                     |                            |                                     |                              |                                       |
|--------------------------------------------------|----------------|--------------|-------------------------|-----------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|------------------------------|---------------------------------------|
| $\ell$                                           | $\gamma(\ell)$ | $\rho(\ell)$ | $\gamma_{\delta}(\ell)$ | $\rho_{\delta}(\ell)$ | $c = 4$                    |                                     | $c = 2$                    |                                     | $c = 1$                    |                                     | $c = 0.5$                    |                                       |
|                                                  |                |              |                         |                       | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}_{\delta}(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_{\delta}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_{\delta}(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_{\delta}(\ell)_{[0.5]}$ |
| Sample size: 2300; sampling interval: 10 seconds |                |              |                         |                       |                            |                                     |                            |                                     |                            |                                     |                              |                                       |
| -4                                               | 0              | 0            |                         |                       | 0.045                      |                                     | 0.042                      |                                     | 0.045                      |                                     | 0.055                        | 0.560                                 |
| -3                                               | 0              | 0            |                         |                       | 0.055                      |                                     | 0.058                      |                                     | 0.058                      |                                     | 0.073                        | 0.577                                 |
| -2                                               | 0.0088         | 0.29         |                         |                       | 0.843                      |                                     | 0.838                      |                                     | 0.854                      |                                     | 0.891                        | 0.580                                 |
| -1                                               | -0.0180        | -0.58        |                         |                       | 1.000                      |                                     | 1.000                      |                                     | 1.000                      |                                     | 1.000                        | 0.724                                 |
| 0                                                | 0.0269         | 0.87         | 0.0217                  | 1.00                  | 1.000                      | 0.998                               | 1.000                      | 0.998                               | 1.000                      | 0.999                               | 1.000                        | 0.932                                 |
| 1                                                | -0.0151        | -0.49        | -0.0115                 | -0.53                 | 0.991                      | 0.990                               | 0.992                      | 0.987                               | 0.993                      | 0.990                               | 0.996                        | 0.698                                 |
| 2                                                | 0.0080         | 0.26         | 0.0061                  | 0.28                  | 0.770                      | 0.775                               | 0.756                      | 0.793                               | 0.779                      | 0.801                               | 0.819                        | 0.596                                 |
| 3                                                | 0              | 0            | 0                       | 0                     | 0.061                      | 0.058                               | 0.058                      | 0.047                               | 0.056                      | 0.053                               | 0.071                        | 0.546                                 |
| 4                                                | 0              | 0            | 0                       | 0                     | 0.047                      | 0.050                               | 0.042                      | 0.047                               | 0.048                      | 0.051                               | 0.056                        | 0.526                                 |
| Sample size: 1550; sampling interval: 15 seconds |                |              |                         |                       |                            |                                     |                            |                                     |                            |                                     |                              |                                       |
| -4                                               | 0              | 0            |                         |                       | 0.037                      |                                     | 0.036                      |                                     | 0.030                      |                                     | 0.054                        | 0.484                                 |
| -3                                               | 0              | 0            |                         |                       | 0.052                      |                                     | 0.042                      |                                     | 0.050                      |                                     | 0.076                        | 0.460                                 |
| -2                                               | 0.0088         | 0.29         |                         |                       | 0.637                      |                                     | 0.624                      |                                     | 0.646                      |                                     | 0.697                        | 0.549                                 |
| -1                                               | -0.0180        | -0.58        |                         |                       | 0.973                      |                                     | 0.981                      |                                     | 0.983                      |                                     | 0.987                        | 0.687                                 |
| 0                                                | 0.0269         | 0.87         | 0.0217                  | 1.00                  | 1.000                      | 0.973                               | 1.000                      | 0.974                               | 1.000                      | 0.983                               | 1.000                        | 0.918                                 |
| 1                                                | -0.0151        | -0.49        | -0.0115                 | -0.53                 | 0.944                      | 0.909                               | 0.952                      | 0.930                               | 0.961                      | 0.921                               | 0.970                        | 0.628                                 |
| 2                                                | 0.0080         | 0.26         | 0.0061                  | 0.28                  | 0.540                      | 0.552                               | 0.554                      | 0.557                               | 0.565                      | 0.597                               | 0.590                        | 0.519                                 |
| 3                                                | 0              | 0            | 0                       | 0                     | 0.046                      | 0.046                               | 0.042                      | 0.044                               | 0.051                      | 0.054                               | 0.071                        | 0.484                                 |
| 4                                                | 0              | 0            | 0                       | 0                     | 0.051                      | 0.047                               | 0.040                      | 0.049                               | 0.046                      | 0.053                               | 0.058                        | 0.486                                 |

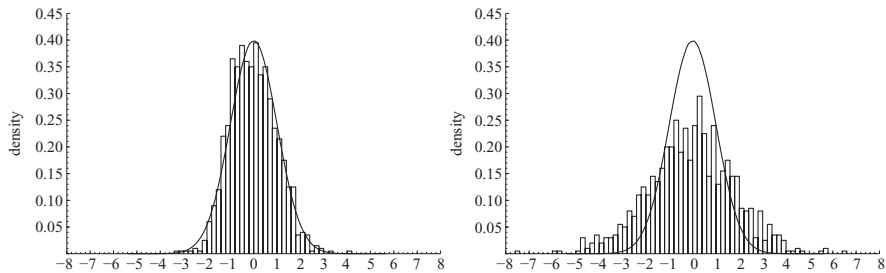
Sample size: 770; sampling interval: 30 seconds

|    |         |       |         |       |       |       |       |       |
|----|---------|-------|---------|-------|-------|-------|-------|-------|
| -4 | 0       | 0     | 0.038   | 0.041 | 0.038 | 0.059 | 0.281 |       |
| -3 | 0       | 0     | 0.046   | 0.044 | 0.052 | 0.071 | 0.271 |       |
| -2 | 0.0088  | 0.29  | 0.316   | 0.308 | 0.324 | 0.363 | 0.333 |       |
| -1 | -0.0180 | -0.58 | 0.859   | 0.864 | 0.870 | 0.886 | 0.537 |       |
| 0  | 0.0269  | 0.87  | 0.0217  | 1.00  | 0.929 | 0.845 | 0.860 | 0.880 |
| 1  | -0.0151 | -0.49 | -0.0115 | -0.53 | 0.794 | 0.765 | 0.820 | 0.808 |
| 2  | 0.0080  | 0.26  | 0.0061  | 0.28  | 0.237 | 0.232 | 0.234 | 0.275 |
| 3  | 0       | 0     | 0       | 0     | 0.047 | 0.048 | 0.040 | 0.068 |
| 4  | 0       | 0     | 0       | 0     | 0.038 | 0.037 | 0.029 | 0.048 |

Sample size: 2300; sampling interval: 30 seconds in 3 days

|    |         |       |         |       |       |       |       |       |       |       |       |       |       |
|----|---------|-------|---------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| -4 | 0       | 0     |         | 0.048 | 0.053 | 0.055 | 0.067 | 0.386 |       |       |       |       |       |
| -3 | 0       | 0     |         | 0.067 | 0.071 | 0.067 | 0.080 | 0.383 |       |       |       |       |       |
| -2 | 0.0088  | 0.29  |         | 0.720 | 0.715 | 0.726 | 0.753 | 0.584 |       |       |       |       |       |
| -1 | -0.0180 | -0.58 |         | 0.939 | 0.945 | 0.951 | 0.956 | 0.845 |       |       |       |       |       |
| 0  | 0.0269  | 0.87  | 0.0217  | 1.00  | 0.939 | 0.944 | 0.997 | 0.951 | 0.955 | 0.986 |       |       |       |
| 1  | -0.0151 | -0.49 | -0.0115 | -0.53 | 0.851 | 0.923 | 0.876 | 0.882 | 0.941 | 0.887 |       |       |       |
| 2  | 0.0080  | 0.26  | 0.0061  | 0.28  | 0.638 | 0.635 | 0.648 | 0.657 | 0.643 | 0.658 |       |       |       |
| 3  | 0       | 0     | 0       | 0     | 0.067 | 0.060 | 0.062 | 0.064 | 0.074 | 0.075 |       |       |       |
| 4  | 0       | 0     | 0       | 0     | 0.039 | 0.058 | 0.035 | 0.055 | 0.040 | 0.053 | 0.046 | 0.068 | 0.362 |

The market microstructure noises are generated by a bivariate MA(2) process.  $\ell = 1$  represents one tick time whose length is equal to each sampling interval (10, 15, and 30 seconds). Columns from second to fifth represent true cross and auto covariances  $\gamma(\ell)$ ,  $\gamma_{\delta}(\ell)$  and true cross and auto correlation coefficients  $\rho(\ell)$ ,  $\rho_{\delta}(\ell)$ .



**Figure 6** Standardized empirical distributions of  $\hat{\gamma}(\ell)$  (left side) and  $\hat{\gamma}(\ell)$  (right side), and the standard normal distributions (solid line).

where  $\mu_y$  is a constant and  $(\mu_1, \mu_2)$  are set to  $(0.1, 0.1)$ ,  $(0.5, 0.5)$ , and  $(1, 1)$ . We estimate  $\hat{\gamma}(\ell)_{[1]}$  and  $\hat{\gamma}_\delta(\ell)_{[1]}$  under the above settings. Table 5 represents the bias and RMSE ratio of  $\hat{\gamma}(\ell)_{[1]}$  and  $\hat{\gamma}_\delta(\ell)_{[1]}$ , estimated under the above setting relative to  $\gamma(\ell)_{[1]}$  and  $\gamma_\delta(\ell)_{[1]}$ , estimated with a data generation process without drift as in (23). The magnitude of the estimation bias is rather small and the RMSE ratios take values that are close to one, even in the case of high values for  $(\mu_1, \mu_2) = (1, 1)$ , which indicate rapid trend acceleration. Thus, the evidence suggests that the noise covariance estimators are not substantially affected by the presence of nonzero drift terms.

**4.1.2 Bivariate autoregressive noise.** Following the above analysis under regular and synchronous sampling, it is important also to consider the irregular and nonsynchronous sampling, which is more representative of actual trading systems. The average observed time intervals for assets 1 and 2 are 10 and 5 seconds, respectively. The estimation is performed with the bivariate autoregressive noise process and the autocorrelation decays gradually to zero, but it is not truncated at some finite lag. Although Assumption (1b) is approximately valid, it is important to investigate the finite sample properties of the noise covariance estimators. Thus, the noise terms  $\eta(t)$  and  $\delta(t)$  are generated by the following bivariate AR(1) model every one second:

$$\begin{aligned}\eta(t) &= 0.6 \eta(t-1) + 0.2 \delta(t-1) + \epsilon_\eta(t), \\ \delta(t) &= 0.2 \eta(t-1) + 0.6 \delta(t-1) + \epsilon_\delta(t).\end{aligned}$$

After summing up the equilibrium prices and noise terms, the observed prices are randomly selected by Poisson sampling with mean durations of 10 and 5 seconds for assets 1 and 2, respectively. In the irregular and nonsynchronous sampling,  $\ell = 1$  represents one second. Under this simulation setting, there are on average, 900 realizations of the sample size  $N_\ell$  with  $\ell = 0$ . On the other hand, the average sample size of  $N_\ell$  with  $|\ell| > 0$  is 450. For given  $c$  values,  $M_\ell$  and  $K_\ell$  amount to  $(c, M_\ell, K_\ell) = (4, 30, 15)$ ,  $(2, 15, 30)$ ,  $(1, 8, 45)$ , and  $(0.5, 4, 110)$ .

**Table 5** The bias and RMSE ratio of  $\hat{\gamma}(\ell)_{[1]}$  and  $\hat{\gamma}_\delta(\ell)_{[1]}$  with drift parameter  $(\mu_1, \mu_2)$  to  $\hat{\gamma}(\ell)_{[1]}$  and  $\hat{\gamma}_\delta(\ell)_{[1]}$  without drift

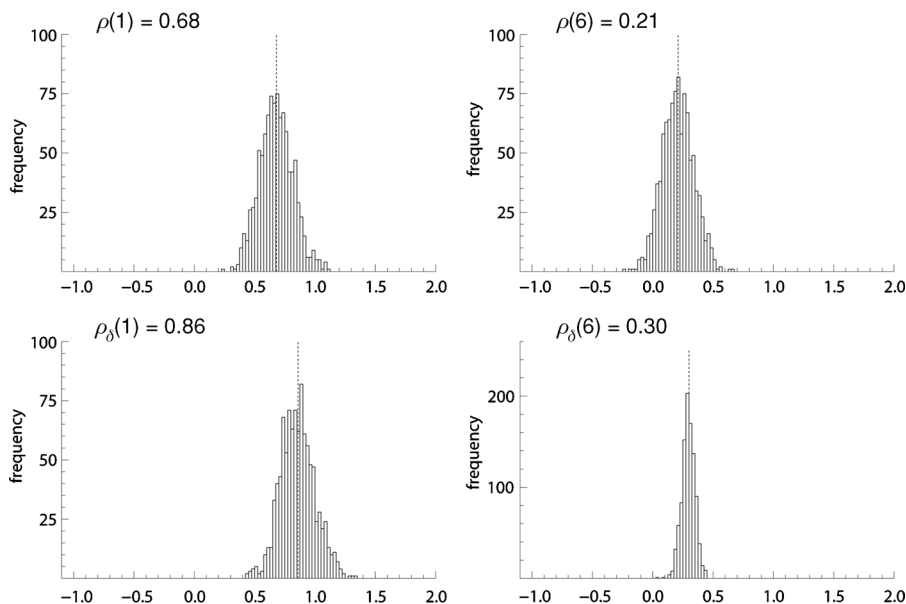
| $\ell$                                           | No drift          |                   |            |  | (0.1,0.1)         |            |                   |            | (0.5,0.5)         |            |                   |            | (1,1)             |            |                   |            |
|--------------------------------------------------|-------------------|-------------------|------------|--|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|-------------------|------------|
|                                                  | bias $\times$ 100 | bias $\times$ 100 | RMSE ratio |  | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio |
| Sample size: 2300; sampling interval: 10 seconds |                   |                   |            |  |                   |            |                   |            |                   |            |                   |            |                   |            |                   |            |
| -4                                               | -0.0155           | -0.0156           | 1.0001     |  | -0.0157           | 1.0004     | -0.0161           |            | -0.0157           | 1.0004     | -0.0161           |            | -0.0161           | 1.0010     | -0.0161           | 1.0010     |
| -3                                               | 0.0321            | 0.0321            | 1.0001     |  | 0.0316            | 0.9993     | 0.0312            |            | 0.0316            | 0.9993     | 0.0312            |            | 0.0312            | 1.0000     | 0.0312            | 1.0000     |
| -2                                               | -0.0594           | -0.0594           | 1.0000     |  | -0.0593           | 0.9989     | -0.0597           |            | -0.0593           | 0.9989     | -0.0597           |            | -0.0597           | 0.9992     | -0.0597           | 0.9992     |
| -1                                               | -0.1001           | -0.1001           | 1.0000     |  | -0.1008           | 0.9998     | -0.1012           |            | -0.1008           | 0.9998     | -0.1012           |            | -0.1012           | 0.9999     | -0.1012           | 0.9999     |
| 0                                                | -0.1798           | -0.1280           | 1.0000     |  | -0.1798           | 1.0001     | -0.1803           |            | -0.1803           | 1.0004     | -0.1809           |            | -0.1809           | 1.0006     | -0.1271           | 0.9977     |
| 1                                                | -0.1944           | -0.1024           | 1.0000     |  | -0.1944           | 1.0001     | -0.1946           |            | -0.1946           | 1.0001     | -0.1951           |            | -0.1951           | 1.0002     | -0.1016           | 0.9976     |
| 2                                                | -0.0675           | -0.0439           | 1.0000     |  | -0.0675           | 1.0000     | -0.0678           |            | -0.0678           | 1.0001     | -0.0682           |            | -0.0682           | 1.0003     | -0.0443           | 0.9998     |
| 3                                                | 0.0347            | 0.0179            | 1.0000     |  | 0.0346            | 1.0002     | 0.0344            |            | 0.0344            | 1.0001     | 0.0340            |            | 0.0340            | 1.0003     | 0.0173            | 1.0030     |
| 4                                                | -0.0167           | -0.0125           | 1.0001     |  | -0.0167           | 1.0002     | -0.0169           |            | -0.0169           | 1.0006     | -0.0174           |            | -0.0174           | 1.0014     | -0.0127           | 1.0030     |
| Sample size: 1550; sampling interval: 15 seconds |                   |                   |            |  |                   |            |                   |            |                   |            |                   |            |                   |            |                   |            |
| -4                                               | -0.0333           | -0.0333           | 0.9999     |  | -0.0335           | 0.9995     | -0.0343           |            | -0.0335           | 0.9995     | -0.0343           |            | -0.0343           | 0.9994     | -0.0343           | 0.9994     |
| -3                                               | 0.0704            | 0.0703            | 0.9998     |  | 0.0701            | 0.9988     | 0.0693            |            | 0.0701            | 0.9988     | 0.0693            |            | 0.0693            | 0.9978     | 0.0693            | 0.9978     |
| -2                                               | -0.1163           | -0.1163           | 1.0001     |  | -0.1165           | 1.0005     | -0.1173           |            | -0.1165           | 1.0005     | -0.1173           |            | -0.1173           | 1.0013     | -0.1173           | 1.0013     |
| -1                                               | -0.3067           | -0.3067           | 0.9999     |  | -0.3070           | 0.9996     | -0.3077           |            | -0.3070           | 0.9996     | -0.3077           |            | -0.3077           | 0.9991     | -0.3077           | 0.9991     |
| 0                                                | -0.3686           | -0.2507           | 0.9999     |  | -0.3686           | 0.9984     | -0.3690           |            | -0.3690           | 0.9998     | -0.3705           |            | -0.3705           | 0.9998     | -0.2522           | 0.9993     |
| 1                                                | -0.3384           | -0.2190           | 1.0000     |  | -0.3385           | 0.9981     | -0.3387           |            | -0.3387           | 0.9998     | -0.3400           |            | -0.3400           | 0.9997     | -0.2199           | 0.9976     |
| 2                                                | -0.1253           | -0.0912           | 1.0000     |  | -0.1253           | 0.9997     | -0.1255           |            | -0.1255           | 1.0003     | -0.1259           |            | -0.1259           | 1.0008     | -0.0923           | 0.9981     |
| 3                                                | 0.0890            | 0.0372            | 0.9999     |  | 0.0890            | 0.9996     | 0.0888            |            | 0.0888            | 0.9998     | 0.0876            |            | 0.0876            | 1.0000     | 0.0365            | 0.9982     |
| 4                                                | -0.0349           | -0.0275           | 0.9998     |  | -0.0349           | 0.9994     | -0.0351           |            | -0.0351           | 0.9992     | -0.0357           |            | -0.0357           | 0.9985     | -0.0286           | 0.9972     |

(continued)

Table 5 (continued)

| $\ell$                                                     | (0.1,0.1)         |                   |            | (0.5,0.5)         |            |                   | (1,1)      |        |
|------------------------------------------------------------|-------------------|-------------------|------------|-------------------|------------|-------------------|------------|--------|
|                                                            | No drift          |                   |            |                   |            |                   |            |        |
|                                                            | bias $\times$ 100 | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio | bias $\times$ 100 | RMSE ratio |        |
| Sample size: 770; sampling interval: 30 seconds            |                   |                   |            |                   |            |                   |            |        |
| -4                                                         | -0.0168           | -0.0168           | 1.0006     | -0.0179           | 1.0031     | -0.0219           | 1.0074     |        |
| -3                                                         | 0.0931            | 0.0931            | 1.0007     | 0.0923            | 1.0040     | 0.0891            | 1.0086     |        |
| -2                                                         | -0.1132           | -0.1133           | 1.0007     | -0.1136           | 1.0040     | -0.1172           | 1.0083     |        |
| -1                                                         | -0.5770           | -0.5770           | 1.0001     | -0.5794           | 1.0005     | -0.5848           | 1.0016     |        |
| 0                                                          | -0.6288           | -0.6288           | 1.0000     | -0.6304           | 1.0002     | -0.6351           | 1.0023     | 1.0061 |
| 1                                                          | -0.5895           | -0.4351           | 0.9999     | -0.5898           | 0.9994     | -0.5918           | 0.9990     | 1.0054 |
| 2                                                          | -0.1517           | -0.0997           | 0.9996     | -0.1519           | 0.9987     | -0.1529           | 0.9962     | 1.0140 |
| 3                                                          | 0.0917            | 0.0489            | 0.9997     | 0.0914            | 0.9986     | 0.0901            | 0.9970     | 1.0148 |
| 4                                                          | -0.0378           | -0.0272           | 0.9996     | -0.0381           | 0.9984     | -0.0380           | 0.9925     | 1.0140 |
| Sample size: 2300; sampling interval: 30 seconds in 3 days |                   |                   |            |                   |            |                   |            |        |
| -4                                                         | -0.0244           | -0.0245           | 1.0003     | -0.0258           | 1.0024     | -0.0292           | 1.0070     |        |
| -3                                                         | 0.0359            | 0.0358            | 1.0003     | 0.0346            | 1.0022     | 0.0312            | 1.0057     |        |
| -2                                                         | -0.0719           | -0.0720           | 0.9999     | -0.0733           | 1.0000     | -0.0767           | 1.0020     |        |
| -1                                                         | -0.1356           | -0.1357           | 1.0002     | -0.1370           | 1.0011     | -0.1404           | 1.0026     |        |
| 0                                                          | -0.2098           | -0.1575           | 1.0002     | -0.2114           | 1.0015     | -0.2167           | 1.0062     | 1.0071 |
| 1                                                          | -0.2076           | -0.1320           | 1.0001     | -0.2087           | 1.0004     | -0.2127           | 1.0037     | 1.0070 |
| 2                                                          | -0.0913           | -0.0488           | 1.0003     | -0.0934           | 1.0025     | -0.0979           | 1.0079     | 1.0050 |
| 3                                                          | 0.0365            | 0.0203            | 1.0004     | 0.0354            | 1.0009     | 0.0305            | 1.0077     | 1.0059 |
| 4                                                          | -0.0269           | -0.0065           | 1.0006     | -0.0291           | 1.0030     | -0.0326           | 1.0090     | 1.0081 |

The market microstructure noises are generated by a bivariate MA(2) process.  $\ell = 1$  represents one tick time whose length is equal to each sampling interval. In the simulation we use the stochastic differential equations with drift parameter  $(\mu_1, \mu_2) = (0.1, 0.1)$ ,  $(0.5, 0.5)$ , and  $(1, 1)$ .



**Figure 7** Empirical distributions of  $\hat{\rho}(\ell)$  and  $\hat{\rho}_\delta(\ell)$

In the simulation for autocovariance estimator of univariate noise process,  $N_{2,\ell}$  for  $\ell = 1, 2$ , which represents the sample size for asset 2, amounts to around  $1/6$  and  $1/2$  of  $N_{2,\ell}$  for  $\ell > 2$  because the average observed time interval for asset 2 is set at 5 seconds. For empirical distribution of  $\hat{\rho}_\delta(\ell)$  is estimated using the sample over a period of three days. Each vertical dashed line represents true correlation value.

Tables 6 and 7 summarize the bias and RMSE of the cross-covariance and autocovariance estimators  $\hat{\gamma}(\ell)_{[c]}$ ,  $\hat{\gamma}_\delta(\ell)_{[c]}$  proposed here as well as the Voev and Lunde's (2007) cross-covariance estimator  $\check{\gamma}(\ell)$ . For the cross-covariance estimators, the bias of  $\hat{\gamma}(\ell)_{[c]}$  and  $\check{\gamma}(\ell)$  is virtually zero, and  $\hat{\gamma}(\ell)_{[c]}$  has a smaller RMSE than  $\check{\gamma}(\ell)$  for all values of  $\ell$ . The smaller RMSE of  $\hat{\gamma}(\ell)_{[c]}$  can be explained by the shorter intervals used for the construction of  $\hat{\gamma}(\ell)_{[c]}$  relative to  $\check{\gamma}(\ell)$ . For the autocovariance estimator, the bias of  $\hat{\gamma}(\ell)_{[c]}$  is close to zero and the RMSE seems to be independent of the value of  $c$ .<sup>1</sup>

Figure 7 shows the empirical distributions of cross-correlation and autocorrelation estimates  $\hat{\rho}(\ell) = \hat{\gamma}(\ell)_{[1]} / \sqrt{\hat{\gamma}_\eta(0)_{[1]} \hat{\gamma}_\delta(0)_{[1]}}$  and  $\hat{\rho}_\delta(\ell) = \hat{\gamma}_\delta(\ell)_{[1]} / \hat{\gamma}_\delta(0)_{[1]}$ , respectively. It is clear that the correlation estimates provide good approximations of the true correlation. The standardized distribution of the cross and auto covariance estimators  $\hat{\gamma}(\ell)_{[c]}$  and  $\hat{\gamma}_\delta(\ell)_{[c]}$  given the value of  $c$  are also presented in Figures 8 and 9. The average sample size within each subseries  $M_\ell$  is 4 when  $c$  is equal to 0.5.

<sup>1</sup>It is noted that the RMSE of  $\hat{\gamma}_\delta(\ell)$  for  $\ell = 1, 2$  takes larger values than those associated with  $\hat{\gamma}_\delta(\ell)$  for  $\ell > 2$ . This is due to the smaller  $N_{2,\ell}$  for  $\ell = 1, 2$ , which represents the sample size for asset 2. In the present numerical example,  $N_{2,1}$  and  $N_{2,2}$  amount to around  $1/6$  and  $1/2$  of  $N_{2,\ell}$  for  $\ell > 2$  because the average observed time interval for asset 2 is set at 5 seconds.

Table 6 Bias of cross and auto covariance estimators of bivariate AR noise processes

| True covariances |                |              |                  |                |                |                            |                              |                            |                              |                            |                              |                              |                                |
|------------------|----------------|--------------|------------------|----------------|----------------|----------------------------|------------------------------|----------------------------|------------------------------|----------------------------|------------------------------|------------------------------|--------------------------------|
| $\ell$           | $\gamma(\ell)$ | $\rho(\ell)$ | $\gamma_s(\ell)$ |                | $\rho_s(\ell)$ | $c = 4$                    |                              | $c = 2$                    |                              | $c = 1$                    |                              | $c = 0.5$                    |                                |
|                  |                |              | $\gamma_s(\ell)$ | $\rho_s(\ell)$ |                | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}_s(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_s(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_s(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_s(\ell)_{[0.5]}$ |
| -30              | 0.0000         | 0.00         |                  |                |                | 0.0000                     |                              | -0.0001                    |                              | -0.0001                    |                              | -0.0001                      | -0.0010                        |
| -25              | 0.0001         | 0.00         |                  |                |                | 0.0001                     |                              | 0.0001                     |                              | 0.0000                     |                              | 0.0000                       | -0.0008                        |
| -20              | 0.0004         | 0.01         |                  |                |                | 0.0000                     |                              | -0.0001                    |                              | -0.0003                    |                              | -0.0001                      | -0.0008                        |
| -15              | 0.0013         | 0.04         |                  |                |                | 0.0000                     |                              | 0.0000                     |                              | 0.0000                     |                              | -0.0001                      | -0.0005                        |
| -10              | 0.0039         | 0.12         |                  |                |                | -0.0003                    |                              | -0.0004                    |                              | -0.0004                    |                              | -0.0004                      | -0.0005                        |
| -9               | 0.0048         | 0.16         |                  |                |                | 0.0000                     |                              | -0.0001                    |                              | -0.0002                    |                              | -0.0003                      | -0.0011                        |
| -8               | 0.0060         | 0.19         |                  |                |                | -0.0003                    |                              | -0.0004                    |                              | -0.0004                    |                              | -0.0005                      | -0.0007                        |
| -7               | 0.0075         | 0.24         |                  |                |                | -0.0001                    |                              | -0.0003                    |                              | -0.0003                    |                              | -0.0004                      | -0.0007                        |
| -6               | 0.0094         | 0.30         |                  |                |                | 0.0001                     |                              | 0.0001                     |                              | -0.0002                    |                              | -0.0001                      | -0.0006                        |
| -5               | 0.0117         | 0.38         |                  |                |                | -0.0001                    |                              | -0.0002                    |                              | -0.0002                    |                              | -0.0002                      | -0.0004                        |
| -4               | 0.0145         | 0.47         |                  |                |                | -0.0002                    |                              | -0.0003                    |                              | -0.0003                    |                              | -0.0004                      | -0.0001                        |
| -3               | 0.0179         | 0.58         |                  |                |                | 0.0000                     |                              | -0.0003                    |                              | -0.0002                    |                              | -0.0003                      | -0.0010                        |
| -2               | 0.0217         | 0.70         |                  |                |                | -0.0001                    |                              | -0.0004                    |                              | -0.0005                    |                              | -0.0005                      | -0.0004                        |
| -1               | 0.0255         | 0.83         |                  |                |                | -0.0001                    |                              | -0.0002                    |                              | -0.0003                    |                              | -0.0003                      | -0.0007                        |
| 0                | 0.0278         | 0.90         | 0.0217           | 1.00           |                | 0.0000                     | -0.0002                      | -0.0002                    | -0.0003                      | -0.0003                    | -0.0004                      | -0.0004                      | -0.0003                        |
| 1                | 0.0210         | 0.68         | 0.0186           | 0.86           |                | -0.0002                    | 0.0001                       | -0.0003                    | 0.0000                       | -0.0004                    | -0.0004                      | 0.0000                       | -0.0005                        |
| 2                | 0.0163         | 0.53         | 0.0154           | 0.71           |                | -0.0001                    | -0.0003                      | -0.0002                    | -0.0005                      | -0.0002                    | -0.0004                      | -0.0006                      | -0.0001                        |
| 3                | 0.0129         | 0.42         | 0.0125           | 0.58           |                | 0.0000                     | -0.0001                      | 0.0000                     | -0.0002                      | 0.0000                     | -0.0002                      | -0.0002                      | -0.0005                        |
| 4                | 0.0102         | 0.33         | 0.0101           | 0.46           |                | -0.0002                    | -0.0001                      | -0.0002                    | -0.0002                      | -0.0002                    | -0.0002                      | -0.0003                      | -0.0004                        |
| 5                | 0.0081         | 0.26         | 0.0081           | 0.37           |                | 0.0001                     | -0.0001                      | 0.0000                     | -0.0001                      | 0.0000                     | -0.0002                      | -0.0002                      | 0.0002                         |
| 6                | 0.0065         | 0.21         | 0.0065           | 0.30           |                | -0.0001                    | -0.0001                      | -0.0002                    | -0.0002                      | -0.0002                    | -0.0002                      | -0.0002                      | -0.0001                        |
| 7                | 0.0052         | 0.17         | 0.0052           | 0.24           |                | -0.0003                    | -0.0002                      | -0.0002                    | -0.0002                      | -0.0002                    | -0.0002                      | -0.0003                      | -0.0005                        |
| 8                | 0.0042         | 0.13         | 0.0042           | 0.19           |                | -0.0003                    | 0.0000                       | -0.0002                    | 0.0000                       | -0.0002                    | 0.0000                       | 0.0000                       | -0.0004                        |
| 9                | 0.0033         | 0.11         | 0.0033           | 0.15           |                | 0.0000                     | -0.0002                      | -0.0002                    | -0.0002                      | -0.0002                    | -0.0002                      | -0.0002                      | -0.0005                        |
| 10               | 0.0027         | 0.09         | 0.0027           | 0.12           |                | -0.0002                    | -0.0001                      | -0.0001                    | -0.0002                      | -0.0001                    | -0.0002                      | -0.0002                      | -0.0006                        |
| 15               | 0.0009         | 0.03         | 0.0009           | 0.04           |                | -0.0003                    | 0.0000                       | -0.0002                    | -0.0001                      | -0.0001                    | 0.0000                       | 0.0000                       | -0.0002                        |
| 20               | 0.0003         | 0.01         | 0.0003           | 0.01           |                | 0.0001                     | 0.0000                       | 0.0001                     | 0.0000                       | 0.0002                     | 0.0000                       | 0.0000                       | -0.0003                        |
| 25               | 0.0001         | 0.00         | 0.0001           | 0.00           |                | 0.0002                     | 0.0000                       | 0.0001                     | 0.0000                       | 0.0002                     | 0.0000                       | 0.0000                       | -0.0004                        |
| 30               | 0.0000         | 0.00         | 0.0000           | 0.00           |                | 0.0001                     | 0.0000                       | 0.0000                     | 0.0000                       | 0.0000                     | 0.0000                       | 0.0000                       | -0.0003                        |

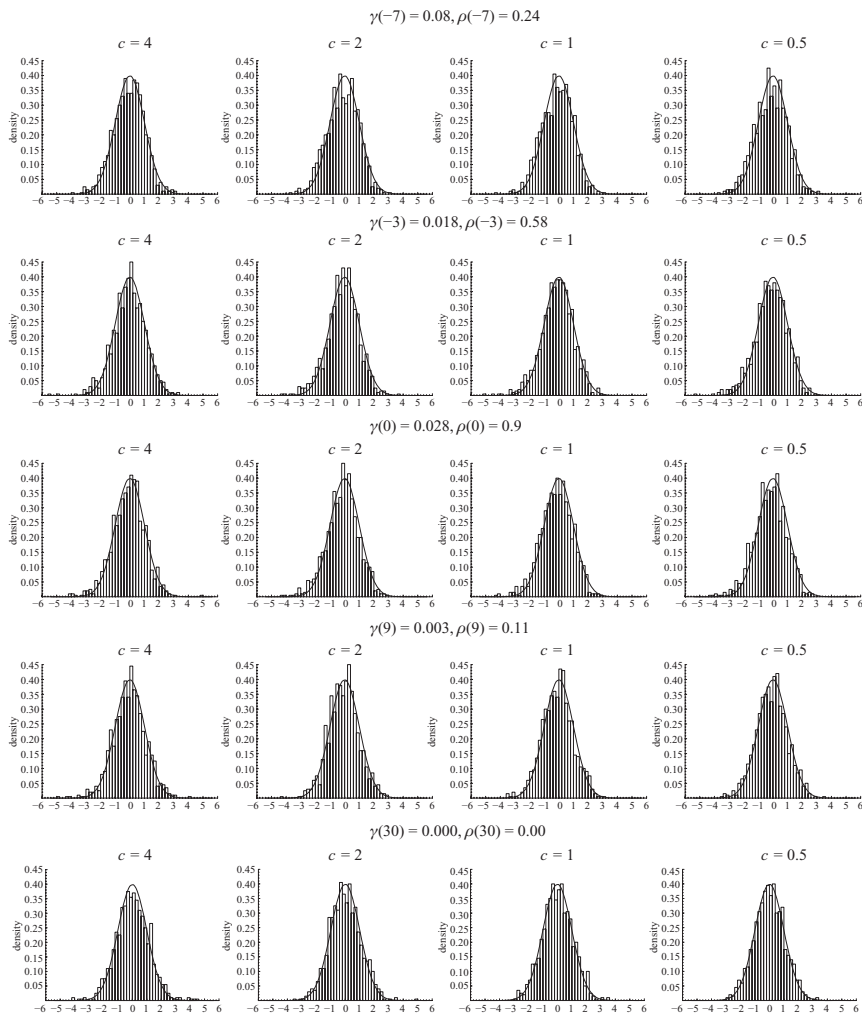
The market microstructure noises are generated by a bivariate AR(1) process. Columns from second to fifth represent true cross and auto covariances  $\gamma(\ell)$ ,  $\gamma_s(\ell)$  and true cross and auto correlation coefficients  $\rho(\ell)$ ,  $\rho_s(\ell)$ .



Table 7 RMSE of cross and auto covariance estimators of bivariate AR noise processes

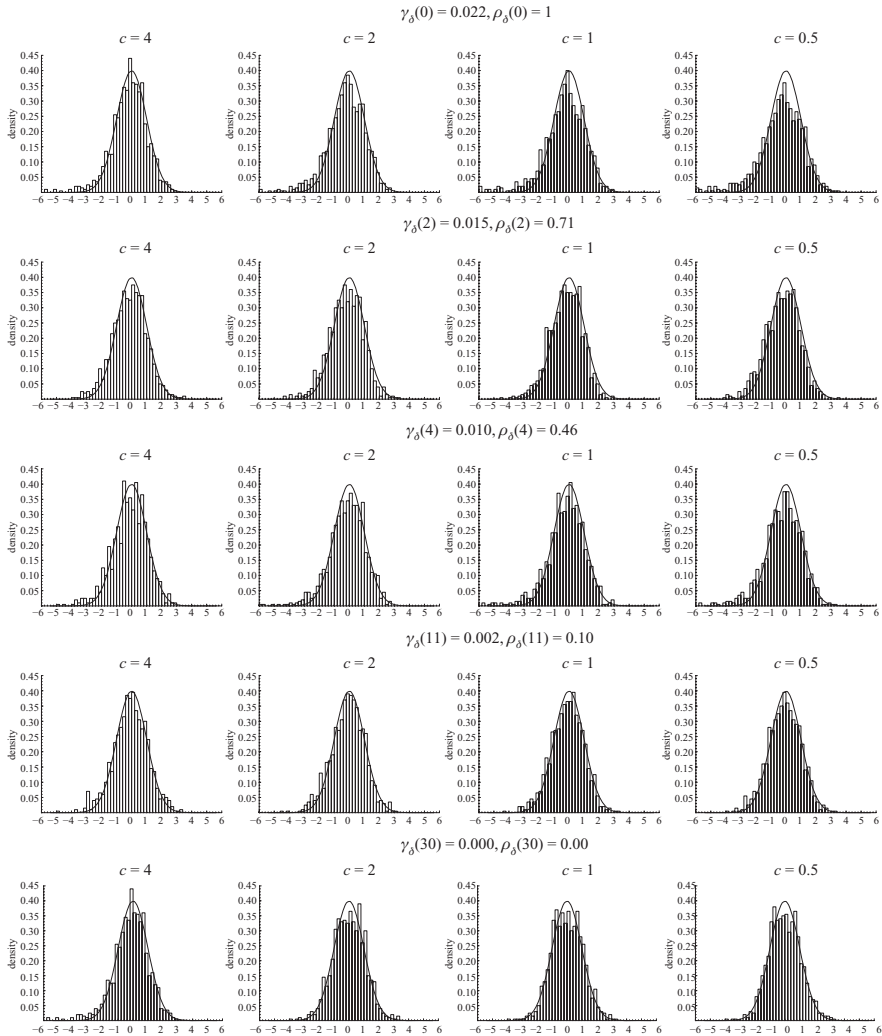
| $\ell$ | True covariances |              |                         |                       | $c = 4$                    |                                     | $c = 2$                    |                                     | $c = 1$                    |                                     | $c = 0.5$                    |                                       | RMSE ratio<br>$\frac{\hat{\gamma}(\ell)}{\hat{\gamma}(\ell)_{[0.5]}}$ |
|--------|------------------|--------------|-------------------------|-----------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|----------------------------|-------------------------------------|------------------------------|---------------------------------------|-----------------------------------------------------------------------|
|        | $\gamma(\ell)$   | $\rho(\ell)$ | $\gamma_{\delta}(\ell)$ | $\rho_{\delta}(\ell)$ | $\hat{\gamma}(\ell)_{[4]}$ | $\hat{\gamma}_{\delta}(\ell)_{[4]}$ | $\hat{\gamma}(\ell)_{[2]}$ | $\hat{\gamma}_{\delta}(\ell)_{[2]}$ | $\hat{\gamma}(\ell)_{[1]}$ | $\hat{\gamma}_{\delta}(\ell)_{[1]}$ | $\hat{\gamma}(\ell)_{[0.5]}$ | $\hat{\gamma}_{\delta}(\ell)_{[0.5]}$ |                                                                       |
| -30    | 0.0000           | 0.00         |                         |                       | 0.0034                     | 0.0034                              | 0.0034                     | 0.0034                              | 0.0033                     | 0.0033                              | 0.0033                       | 0.0033                                | 5.6022                                                                |
| -25    | 0.0001           | 0.00         |                         |                       | 0.0036                     | 0.0036                              | 0.0035                     | 0.0035                              | 0.0035                     | 0.0035                              | 0.0034                       | 0.0034                                | 5.3962                                                                |
| -20    | 0.0004           | 0.01         |                         |                       | 0.0037                     | 0.0037                              | 0.0037                     | 0.0037                              | 0.0037                     | 0.0037                              | 0.0035                       | 0.0035                                | 5.0531                                                                |
| -15    | 0.0013           | 0.04         |                         |                       | 0.0038                     | 0.0038                              | 0.0038                     | 0.0038                              | 0.0038                     | 0.0038                              | 0.0038                       | 0.0038                                | 4.7665                                                                |
| -10    | 0.0039           | 0.12         |                         |                       | 0.0039                     | 0.0039                              | 0.0038                     | 0.0038                              | 0.0038                     | 0.0038                              | 0.0039                       | 0.0039                                | 4.8153                                                                |
| -9     | 0.0048           | 0.16         |                         |                       | 0.0039                     | 0.0039                              | 0.0039                     | 0.0039                              | 0.0039                     | 0.0039                              | 0.0038                       | 0.0038                                | 4.6998                                                                |
| -8     | 0.0060           | 0.19         |                         |                       | 0.0039                     | 0.0039                              | 0.0040                     | 0.0040                              | 0.0040                     | 0.0040                              | 0.0039                       | 0.0039                                | 4.6768                                                                |
| -7     | 0.0075           | 0.24         |                         |                       | 0.0039                     | 0.0039                              | 0.0040                     | 0.0040                              | 0.0039                     | 0.0039                              | 0.0040                       | 0.0040                                | 4.7858                                                                |
| -6     | 0.0094           | 0.30         |                         |                       | 0.0042                     | 0.0042                              | 0.0042                     | 0.0042                              | 0.0042                     | 0.0042                              | 0.0041                       | 0.0041                                | 4.3739                                                                |
| -5     | 0.0117           | 0.38         |                         |                       | 0.0040                     | 0.0040                              | 0.0039                     | 0.0039                              | 0.0040                     | 0.0040                              | 0.0040                       | 0.0040                                | 4.4678                                                                |
| -4     | 0.0145           | 0.47         |                         |                       | 0.0042                     | 0.0042                              | 0.0043                     | 0.0043                              | 0.0043                     | 0.0043                              | 0.0043                       | 0.0043                                | 4.3749                                                                |
| -3     | 0.0179           | 0.58         |                         |                       | 0.0042                     | 0.0042                              | 0.0041                     | 0.0041                              | 0.0042                     | 0.0042                              | 0.0041                       | 0.0041                                | 4.4923                                                                |
| -2     | 0.0217           | 0.70         |                         |                       | 0.0041                     | 0.0041                              | 0.0041                     | 0.0041                              | 0.0042                     | 0.0042                              | 0.0043                       | 0.0043                                | 4.4703                                                                |
| -1     | 0.0255           | 0.83         |                         |                       | 0.0044                     | 0.0044                              | 0.0046                     | 0.0046                              | 0.0045                     | 0.0045                              | 0.0046                       | 0.0046                                | 4.1387                                                                |
| 0      | 0.0278           | 0.90         | 0.0217                  | 1.00                  | 0.0036                     | 0.0018                              | 0.0036                     | 0.0019                              | 0.0035                     | 0.0018                              | 0.0036                       | 0.0019                                | 3.8068                                                                |
| 1      | 0.0210           | 0.68         | 0.0186                  | 0.86                  | 0.0043                     | 0.0059                              | 0.0044                     | 0.0059                              | 0.0043                     | 0.0057                              | 0.0044                       | 0.0057                                | 4.1603                                                                |
| 2      | 0.0163           | 0.53         | 0.0154                  | 0.71                  | 0.0043                     | 0.0037                              | 0.0043                     | 0.0038                              | 0.0041                     | 0.0036                              | 0.0041                       | 0.0037                                | 4.4145                                                                |
| 3      | 0.0129           | 0.42         | 0.0125                  | 0.58                  | 0.0040                     | 0.0029                              | 0.0041                     | 0.0029                              | 0.0040                     | 0.0029                              | 0.0041                       | 0.0029                                | 4.6493                                                                |
| 4      | 0.0102           | 0.33         | 0.0101                  | 0.46                  | 0.0040                     | 0.0026                              | 0.0039                     | 0.0027                              | 0.0039                     | 0.0026                              | 0.0040                       | 0.0026                                | 4.6939                                                                |
| 5      | 0.0081           | 0.26         | 0.0081                  | 0.37                  | 0.0041                     | 0.0025                              | 0.0041                     | 0.0025                              | 0.0041                     | 0.0025                              | 0.0040                       | 0.0025                                | 4.4672                                                                |
| 6      | 0.0065           | 0.21         | 0.0065                  | 0.30                  | 0.0041                     | 0.0023                              | 0.0039                     | 0.0024                              | 0.0040                     | 0.0023                              | 0.0039                       | 0.0023                                | 4.5523                                                                |
| 7      | 0.0052           | 0.17         | 0.0052                  | 0.24                  | 0.0041                     | 0.0024                              | 0.0041                     | 0.0024                              | 0.0041                     | 0.0023                              | 0.0041                       | 0.0023                                | 4.4815                                                                |
| 8      | 0.0042           | 0.13         | 0.0042                  | 0.19                  | 0.0040                     | 0.0024                              | 0.0040                     | 0.0024                              | 0.0039                     | 0.0024                              | 0.0039                       | 0.0024                                | 4.5172                                                                |
| 9      | 0.0033           | 0.11         | 0.0033                  | 0.15                  | 0.0039                     | 0.0024                              | 0.0038                     | 0.0024                              | 0.0039                     | 0.0024                              | 0.0039                       | 0.0024                                | 4.6349                                                                |
| 10     | 0.0027           | 0.09         | 0.0027                  | 0.12                  | 0.0039                     | 0.0024                              | 0.0040                     | 0.0024                              | 0.0039                     | 0.0024                              | 0.0038                       | 0.0023                                | 4.7445                                                                |
| 15     | 0.0009           | 0.03         | 0.0009                  | 0.04                  | 0.0038                     | 0.0022                              | 0.0037                     | 0.0023                              | 0.0036                     | 0.0022                              | 0.0036                       | 0.0022                                | 5.1160                                                                |
| 20     | 0.0003           | 0.01         | 0.0003                  | 0.01                  | 0.0037                     | 0.0021                              | 0.0037                     | 0.0020                              | 0.0036                     | 0.0020                              | 0.0036                       | 0.0020                                | 5.0089                                                                |
| 25     | 0.0001           | 0.00         | 0.0001                  | 0.00                  | 0.0035                     | 0.0020                              | 0.0034                     | 0.0020                              | 0.0033                     | 0.0020                              | 0.0033                       | 0.0019                                | 5.5534                                                                |
| 30     | 0.0000           | 0.00         | 0.0000                  | 0.00                  | 0.0034                     | 0.0019                              | 0.0034                     | 0.0018                              | 0.0033                     | 0.0018                              | 0.0033                       | 0.0018                                | 5.2596                                                                |

The market microstructure noises are generated by a bivariate AR(1) process. Columns from second to fifth represent true cross and auto covariances  $\gamma(\ell)$ ,  $\gamma_{\delta}(\ell)$  and true cross and auto correlation coefficients  $\rho(\ell)$ ,  $\rho_{\delta}(\ell)$ .



**Figure 8** Standardized empirical distributions of the cross-covariance estimator of noises (histogram) and the standard normal distributions (solid line).

The evidence suggests that some standardized empirical distributions are heavy tailed due to smaller  $M_\ell$  and that they approach the standard normal as  $c$  increases. It is thus important that the value of  $c$  in the following empirical application is chosen such that  $M_\ell$  is not too small. It is, however, possible to approximate many observed histograms with the standard normal distribution. Therefore, there is evidence from these simulation results that the estimation and testing procedures are conducive to statistically more reliable covariance estimators. These results are also robust to changes in  $c$  even in the case of autoregressive noise processes.



**Figure 9** Standardized empirical distributions of the autocovariance estimator of noises (histogram) and the standard normal distributions (solid line).

## 4.2 Empirical Illustration

After the above examination of the new covariance estimators (11) and (19) for bivariate noise processes, test statistics (9) and (16) to determine the threshold values and tests of significance of noise cross and auto covariances (14), (21), (22), it is useful to provide the details for implementing these estimation and testing procedures. The estimation and testing for dependence in microstructure noise includes the following steps:

- **Univariate noise process**

- **Threshold value step:** The threshold value  $m_1$  for the univariate noise dependence is determined using the test statistic (16).
- **Estimation step:** The noise variance and autocovariance  $\gamma_\eta(\ell)$  with  $\ell \geq 0$  are estimated using (19).
- **Test significance step:** The null hypotheses  $\gamma_\eta(0) = 0$  and  $\gamma_\eta(\ell) = 0$  are tested using by (21) and (22), respectively.

- **Bivariate noise processes**

- **Threshold value step:** The threshold values of cross-sectional noise dependence  $m^+$  and  $m^-$  are determined using the test statistic (9).
- **Estimation step:** The noise cross-covariance  $\gamma(\ell)$  is estimated using (11).
- **Test significance step:** The null hypothesis  $\gamma(\ell) = 0$  is tested using (14).

These procedures are applied to high-frequency transaction prices of four stocks traded on the Osaka Securities Exchange: OMRON Corporation (OC) and Murata Manufacturing Co., Ltd. (MM) from the manufacturing industry; Ono Pharmaceutical Co., Ltd. (OP) and Santen Pharmaceutical Co., Ltd. (SP) from the medical services industry. The trading hours on the Osaka Securities Exchange are divided between the morning session (9:00–11:00 local time) and the afternoon session (12:30–15:10). As illustrated in Andersen, Bollerslev, and Cai (2000), transaction prices at the opening and closing times are more volatile than any other transaction prices. This is explained by the call auction system to determine the opening and closing prices. In this empirical illustration, transaction prices from 9:05 to 15:05 are used to estimate the noise cross and auto covariances.

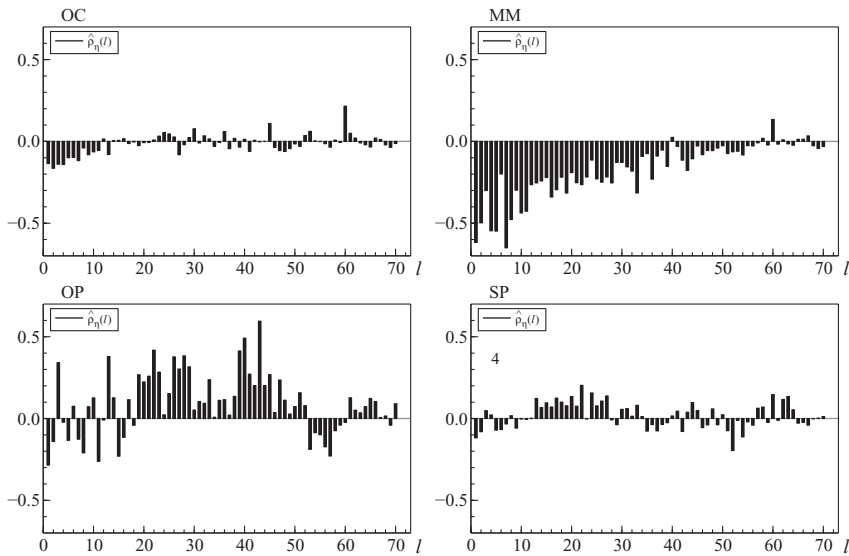
With respect to the univariate noise process, the first step is to test whether  $E_{ij}[Z_{1,L,k}] = 0$  or  $E_{ij}[Z_{1,L,k}] \neq 0$  for all  $k$  with a large value  $L$ , using the test statistic (16) to identify  $m_1$  as described in Section 3.2. When the total number of  $\{Z_{1,\ell,k}\}$  for each  $\ell$  is not large enough for inference purposes, recourse is made to observations over multiple days in order to increase the sample size. In the case of high-frequency data for  $D$  days, the total number of  $\{Z_{1,\ell,k}\}$  on the  $d$ th day where  $d = \{1, \dots, D\}$  is denoted as  $n_\ell^{(d)}$ . For the sample period from October 2, 2006 to September 28, 2007, the number of days is  $D = 245$ . The observations on the first and second day are represented by  $\{Z_{1,\ell,1}, \dots, Z_{1,\ell,n_\ell^{(1)}}\}$  and  $\{Z_{1,\ell,n_\ell^{(1)}+1}, \dots, Z_{1,\ell,n_\ell^{(1)}+n_\ell^{(2)}}\}$ , respectively. The total number of  $\{Z_{1,\ell,k}\}$  in the sample period is  $N_{1,\ell} = \sum_{d=1}^D n_\ell^{(d)}$ . With  $L = 70$  seconds, the null  $E_{ij}[Z_{1,70,k}] = 0$  is tested and if it cannot be rejected, then the null  $E_{ij}[Z_{1,69,k}] = 0$  with lower  $L$  is subsequently tested. This iterative test is applied repetitively until the null is rejected. Finally, the distance for which the null is rejected at first occurrence is considered to represent the threshold value  $m_1$ .

Next, the noise variance  $\gamma_\eta(0)$  is estimated using (19) and the null hypothesis  $\gamma_\eta(0) = 0$  is tested using (21). If the test statistic (21) rejects the null, then the noise autocovariance  $\gamma_\eta(\ell)$  is estimated following (19) and its significance at the 0.05 level is assessed using the test statistic defined in (22). Table 8 (a) reports the estimates of the noise variance and associated test statistics. It presents also the noise-to-signal ratio, which is the ratio of the noise variance to the integrated variance using the kernel-based estimators of Barndorff-Nielsen et al. (2008). The

**Table 8** Autocovariance of univariate noise process

| (a) The variance estimates and the test statistics for the variance |        |         |                       |                       |                       |                       |        |         |
|---------------------------------------------------------------------|--------|---------|-----------------------|-----------------------|-----------------------|-----------------------|--------|---------|
| Industry                                                            |        |         | Manufacturers         |                       | Medicals              |                       |        |         |
|                                                                     |        |         | OC                    | MM                    | OP                    | SP                    |        |         |
| Noise variance                                                      |        |         | $1.17 \times 10^{-6}$ | $1.37 \times 10^{-7}$ | $2.54 \times 10^{-7}$ | $1.04 \times 10^{-6}$ |        |         |
| Test statistics                                                     |        |         | 2240.84*              | 159.09*               | 142.27*               | 1251.67*              |        |         |
| NSR                                                                 |        |         | 0.0049                | 0.0009                | 0.0014                | 0.0042                |        |         |
| (b) The test statistics for the autocovariance                      |        |         |                       |                       |                       |                       |        |         |
| $\ell$                                                              | OC     |         | MM                    |                       | OP                    |                       | SP     |         |
| 1                                                                   | -3.22* | (25132) | -5.00*                | (43753)               | -1.19                 | (14838)               | -1.99* | (12163) |
| 2                                                                   | -3.11* | (15821) | -3.55*                | (31658)               | -0.46                 | (8542)                | -0.95  | (6936)  |
| 3                                                                   | -2.71* | (14909) | -1.52                 | (29671)               | 1.07                  | (7227)                | 0.30   | (5813)  |
| 4                                                                   | -2.58* | (14696) | -3.66*                | (28661)               | -0.08                 | (6415)                | 0.24   | (5677)  |
| 5                                                                   | -1.92  | (12958) | -3.18*                | (27087)               | -0.35                 | (5680)                | -0.83  | (4855)  |
| 6                                                                   | -2.04* | (12599) | -0.99                 | (27243)               | 0.21                  | (5567)                | -0.72  | (4786)  |
| 7                                                                   | -2.00* | (11752) | -4.02*                | (25902)               | -0.45                 | (5130)                | -0.37  | (4385)  |
| 8                                                                   | -0.68  | (11599) | -2.60*                | (25476)               | -0.63                 | (5125)                | 0.25   | (4390)  |
| 9                                                                   | -1.52  | (10923) | -1.63                 | (24362)               | 0.22                  | (4855)                | -0.71  | (3956)  |
| 10                                                                  | -1.36  | (11018) | -2.88*                | (23916)               | 0.51                  | (4879)                | -0.03  | (3960)  |
| 20                                                                  | -0.14  | (9123)  | -1.54                 | (21664)               | 0.72                  | (3643)                | 1.46   | (3223)  |
| 22                                                                  | 0.17   | (8870)  | -2.10*                | (21306)               | 1.46                  | (3432)                | 2.51*  | (3032)  |
| 24                                                                  | 1.16   | (8726)  | -0.65                 | (21264)               | 0.13                  | (3476)                | 2.60*  | (3048)  |
| 30                                                                  | 1.74   | (9672)  | -1.07                 | (21414)               | 0.23                  | (3730)                | 0.76   | (3581)  |
| 40                                                                  | 0.33   | (8284)  | 0.23                  | (20047)               | 1.81                  | (3072)                | 0.22   | (2690)  |
| 43                                                                  | -0.04  | (8148)  | -2.15*                | (19972)               | 3.05*                 | (3077)                | 0.63   | (2749)  |
| 50                                                                  | -0.45  | (8211)  | -0.50                 | (19996)               | 0.56                  | (2983)                | 0.39   | (2775)  |
| 52                                                                  | 1.00   | (8101)  | -1.10                 | (19868)               | 0.63                  | (2879)                | -3.14* | (2666)  |
| 54                                                                  | 0.16   | (8093)  | -1.83                 | (19886)               | -0.70                 | (2927)                | -2.00* | (2798)  |
| 55                                                                  | -0.02  | (7985)  | -0.54                 | (19890)               | -0.85                 | (2903)                | -0.40  | (2582)  |
| 60                                                                  | 5.79*  | (9709)  | 4.89*                 | (21240)               | -0.43                 | (3521)                | 3.19*  | (3545)  |
| 61                                                                  | 1.71   | (8705)  | -0.59                 | (20239)               | 1.96*                 | (3273)                | -0.20  | (3018)  |
| 62                                                                  | 0.82   | (8497)  | 0.31                  | (19754)               | 0.68                  | (3009)                | 2.52*  | (2823)  |
| 63                                                                  | -0.36  | (8336)  | -0.53                 | (19989)               | 0.55                  | (2911)                | 2.81*  | (2843)  |
| 70                                                                  | -0.45  | (7989)  | -1.06                 | (19547)               | 1.24                  | (2850)                | 0.22   | (2598)  |

In part (a), the test statistic for the variance of noise is given by (21). The critical value at 5% significance level is equal to 3.84. NSR is an abbreviation for noise-to-signal ratio, which is the ratio between the variance estimate of the noise and the integrated variance estimate using a realized kernel-based estimator of Barndorff-Nielsen et al. (2008). The part (b) shows the test statistic (22) for the significance of the autocovariance of noise with  $\ell > 0$ . The critical value of the test statistic (22) is 1.96 at 5% significance level. Superscript \* denotes significance at the 5% level. The numbers in parentheses represent the sample size  $N_{1,\ell}$ .



**Figure 10** Autocorrelation functions of the noise in OC, MM, OP, and SP.

results indicate that the variance of market microstructure noise in each asset is significantly larger than zero. Table 8 (b) presents the test statistics for the significance of noise autocovariance following (22). The noise dependence in OC, MM, OP, and SP prices disappears after approximately 60 seconds. Figure 10 plots the autocorrelation functions of microstructure noise  $\hat{\rho}_\eta(l) = \hat{\gamma}_\eta(l)/\hat{\gamma}_\eta(0)$ . The negative autocorrelation at small lags is suggestive of order reversals, where the prevailing buying (selling) orders are likely to be shortly followed by selling (buying) orders. On the other hand, the positive autocorrelation at higher lags is indicative of the clustering of order flows, which occurs in the case of buying or selling pressures. These results are consistent with the evidence from Bandi and Russell (2006).

For the bivariate noise processes, the test statistics defined in (9) are used to determine  $m^+$  and  $m^-$ , and subsequently estimate the cross-covariance by (11). The significance of the cross-covariance is assessed using the test statistics in (14). The results are reported in Table 9. Whether within or across industries, there exist cross-covariances significantly different from zero, including  $-33 \leq \ell \leq 66$  for OC-MM,  $-57 \leq \ell \leq 47$  for OC-OP,  $-62 \leq \ell \leq 0$  for OC-SP,  $-48 \leq \ell \leq 68$  for MM-OP,  $-34 \leq \ell \leq 68$  for MM-SP, and  $-59 \leq \ell \leq 42$  for OP-SP. Figure 11 plots the six cross-correlation functions  $\hat{\rho}(\ell) = \hat{\gamma}(\ell)/\sqrt{\hat{\gamma}_\eta(0)\hat{\gamma}_\delta(0)}$ . The market microstructure noise among some assets is associated with asymmetric and different patterns of cross-sectional dependence. In particular, the asymmetric cross-correlation in OC and SP is reflective of the lead-lag relationship. These empirical results provide

**Table 9** Test statistics for the cross-covariance of bivariate noise processes

| $\ell$ | OC-MM             | OC-OP            | OC-SP            | MM-OP             | MM-SP            | OP-SP            |
|--------|-------------------|------------------|------------------|-------------------|------------------|------------------|
| -70    | -0.91<br>(10615)  | 1.30<br>(4217)   | 0.74<br>(3952)   | -0.55<br>(6527)   | -0.36<br>(6114)  | 0.89<br>(2366)   |
| -62    | -0.50<br>(10801)  | 1.07<br>(4245)   | -2.72*<br>(4105) | -1.06<br>(6500)   | -0.56<br>(6124)  | 0.53<br>(2360)   |
| -59    | 1.01<br>(10987)   | 1.28<br>(4207)   | 1.13<br>(4043)   | -0.87<br>(6467)   | 0.80<br>(6006)   | 2.48*<br>(2456)  |
| -57    | 0.19<br>(11071)   | 2.25*<br>(4092)  | -0.30<br>(3939)  | -0.72<br>(6528)   | -0.70<br>(6161)  | 0.82<br>(2478)   |
| -55    | -0.30<br>(10966)  | 0.90<br>(4143)   | 2.18*<br>(3863)  | -0.68<br>(6355)   | -0.80<br>(6107)  | -0.52<br>(2417)  |
| -48    | -1.60<br>(10905)  | -1.26<br>(4171)  | -0.57<br>(3825)  | 2.51*<br>(6500)   | 0.74<br>(6283)   | -0.68<br>(2455)  |
| -34    | 0.93<br>(10832)   | -1.32<br>(4205)  | -0.88<br>(3889)  | -0.96<br>(6572)   | -2.22*<br>(6213) | -0.07<br>(2442)  |
| -33    | -2.68*<br>(10826) | -0.79<br>(4317)  | 0.67<br>(3874)   | -0.28<br>(6496)   | -1.06<br>(6166)  | 0.60<br>(2386)   |
| -30    | 0.92<br>(10959)   | -0.05<br>(4225)  | -1.06<br>(3959)  | -0.44<br>(6562)   | -0.59<br>(6306)  | 1.47<br>(2506)   |
| -25    | -0.95<br>(10917)  | -1.25<br>(4251)  | -0.88<br>(4051)  | 0.80<br>(6464)    | 1.56<br>(6191)   | -1.06<br>(2442)  |
| -17    | -0.13<br>(10862)  | -0.19<br>(4290)  | -0.29<br>(4069)  | -2.36*<br>(6634)  | 0.31<br>(6283)   | -0.38<br>(2448)  |
| -10    | -1.71<br>(11441)  | -1.16<br>(4398)  | -0.77<br>(4016)  | -1.74<br>(6868)   | -0.72<br>(6431)  | -0.91<br>(2547)  |
| -9     | -2.62*<br>(11197) | -2.47*<br>(4335) | -1.12<br>(4225)  | -1.98*<br>(6863)  | -0.79<br>(6549)  | -1.49<br>(2649)  |
| -8     | -1.85<br>(11214)  | -0.87<br>(4506)  | -1.74<br>(4096)  | -2.44*<br>(7025)  | -0.51<br>(6474)  | -1.52<br>(2668)  |
| -7     | -2.05*<br>(11295) | -1.09<br>(4528)  | -0.03<br>(4108)  | -1.97*<br>(6993)  | -1.62<br>(6556)  | 0.02<br>(2663)   |
| -6     | -3.19*<br>(11457) | -1.35<br>(4575)  | -2.25*<br>(4082) | -2.86*<br>(7158)  | -1.82<br>(6564)  | -0.57<br>(2616)  |
| -5     | -1.45<br>(11411)  | -0.47<br>(4627)  | -2.53*<br>(4129) | -3.88*<br>(7121)  | -1.45<br>(6512)  | -0.34<br>(2706)  |
| -4     | -2.72*<br>(11358) | -0.19<br>(4703)  | -1.34<br>(4200)  | -2.90*<br>(7129)  | -2.90*<br>(6659) | -2.23*<br>(2824) |
| -3     | -4.29*<br>(11577) | -1.01<br>(4776)  | -1.13<br>(4404)  | -2.97*<br>(7300)  | -2.17*<br>(6800) | -1.87<br>(2894)  |
| -2     | -1.90<br>(11671)  | -1.25<br>(4808)  | -0.88<br>(4474)  | -2.96*<br>(7410)  | -2.96*<br>(6739) | -2.53*<br>(2980) |
| -1     | -2.52*<br>(11942) | -1.49<br>(5019)  | -0.53<br>(4657)  | -2.46*<br>(7824)  | -0.78<br>(6948)  | -0.41<br>(3414)  |
| 0      | -2.32*<br>(27263) | -0.21<br>(12576) | 3.58*<br>(11042) | -4.51*<br>(17583) | -0.81<br>(15319) | 2.03*<br>(9256)  |
| 1      | -3.37*<br>(12401) | -1.39<br>(5239)  | -0.67<br>(4751)  | -2.98*<br>(7648)  | -0.67<br>(6822)  | -1.54<br>(3257)  |
| 2      | -2.64*<br>(12046) | -1.94<br>(4832)  | 0.07<br>(4461)   | -3.56*<br>(7391)  | -1.54<br>(6787)  | -1.54<br>(2922)  |
| 3      | -3.19*<br>(11628) | -1.67<br>(4712)  | -0.68<br>(4437)  | -4.81*<br>(7149)  | -1.33<br>(6681)  | -1.32<br>(2807)  |

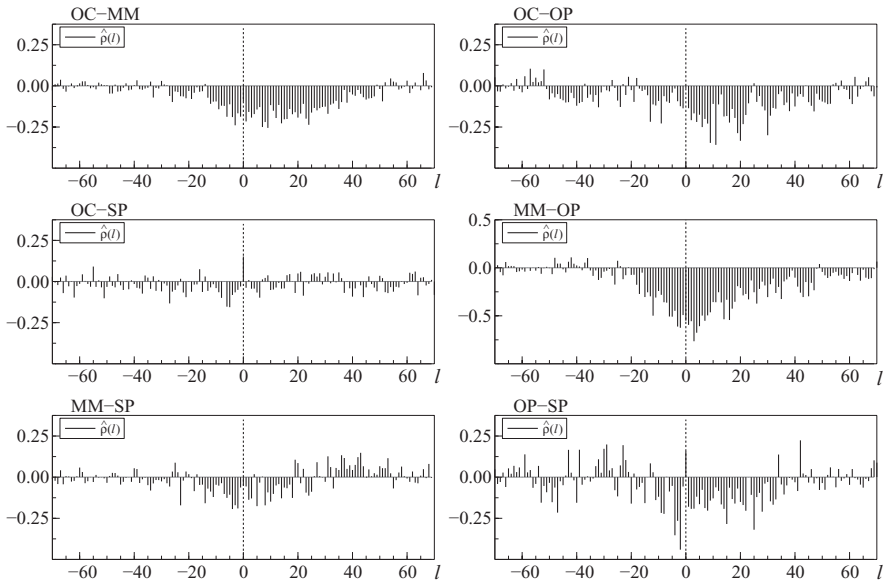
(continued)

Table 9 (continued)

| $\ell$ | OC-MM          | OC-OP         | OC-SP        | MM-OP         | MM-SP         | OP-SP         |
|--------|----------------|---------------|--------------|---------------|---------------|---------------|
| 4      | -2.99* (11473) | -2.03* (4790) | -0.78 (4245) | -4.39* (7141) | 0.24 (6637)   | -1.12 (2853)  |
| 5      | -2.17* (11454) | -1.61 (4619)  | -1.37 (4235) | -3.23* (7197) | -2.05* (6659) | -1.64 (2773)  |
| 6      | -1.78 (11371)  | -2.08* (4496) | -1.78 (4073) | -3.30* (7154) | -0.34 (6568)  | -0.73 (2647)  |
| 7      | -3.87* (11308) | -1.96 (4554)  | 0.24 (4026)  | -3.24* (7081) | 0.03 (6547)   | -1.62 (2613)  |
| 8      | -3.54* (11486) | -2.27* (4602) | 0.38 (4141)  | -3.16* (6987) | -2.14* (6554) | -1.91 (2572)  |
| 9      | -4.08* (11307) | -3.57* (4541) | 0.72 (4089)  | -2.93* (6975) | -1.02 (6233)  | -1.42 (2640)  |
| 10     | -1.87 (11247)  | -1.13 (4399)  | -0.97 (4205) | -2.32* (6770) | -1.75 (6362)  | -1.31 (2551)  |
| 15     | -2.98* (11265) | -2.17* (4373) | -0.73 (4168) | -2.45* (6874) | -0.85 (6347)  | -3.34* (2459) |
| 20     | -2.83* (11053) | -2.97* (4368) | 0.89 (4049)  | -1.35 (6556)  | 1.19 (6300)   | -1.53 (2419)  |
| 30     | -1.98* (11014) | -3.06* (4281) | 0.53 (4024)  | -2.18* (6700) | -0.44 (6251)  | -0.30 (2382)  |
| 42     | -0.48 (10801)  | -0.68 (4206)  | -0.02 (3961) | -2.16* (6531) | 1.91 (6247)   | 4.18* (2345)  |
| 43     | -1.04 (10696)  | -0.99 (4187)  | -0.42 (3981) | -2.77* (6462) | 2.28* (6156)  | 0.44 (2369)   |
| 47     | -1.55 (10989)  | -2.25* (4116) | -0.29 (3897) | -1.93 (6547)  | 0.11 (6064)   | -0.66 (2336)  |
| 64     | -0.05 (10916)  | -0.47 (4225)  | -1.83 (3913) | -2.39* (6398) | -0.91 (6073)  | -0.65 (2337)  |
| 66     | 2.86* (10848)  | 0.61 (4067)   | 0.63 (3996)  | -1.86 (6341)  | 1.19 (6108)   | -0.10 (2424)  |
| 68     | -0.67 (10927)  | -0.72 (4127)  | -0.39 (4011) | -2.16* (6560) | 2.04* (6092)  | 1.04 (2372)   |
| 70     | 0.04 (10733)   | -0.24 (4146)  | -1.25 (3922) | 1.73 (6445)   | -0.02 (6140)  | 1.63 (2381)   |

The test statistic for the significance of the cross-covariance of the bivariate noise processes is given by (14). The critical value of the test statistic (14) is 1.96 at 5% significance level. Superscript \* denotes significance at the 5% level. The numbers in parentheses represent a sample size  $N_\ell$ .





**Figure 11** Cross-correlation functions of the noises among the four assets.

further evidence that the new test statistics and covariance estimators are useful in shedding more light on the market microstructure of noise.

## 5 CONCLUSION

This study provides a method to analyze the market microstructure noise, which occurs due to trading imperfections, frictions, and informational effects. It proposes unbiased and consistent estimators of cross and auto covariances of microstructure noise and derives the related asymptotic distributions. Using Monte Carlo simulation, there is evidence that the proposed cross-covariance estimator is associated with smaller mean squared errors. The larger mean squared errors based on Voev and Lunde (2007) may be induced by the wider time intervals used for the cross-covariance estimation, which are conducive to larger standard deviations. The new estimators are based on time intervals for which the noise dependence disappears. The appropriate length is determined according to a new testing procedure proposed in this paper. Furthermore, the tests of significance of cross and auto covariances are also defined. The simulation analysis suggests that these test statistics are associated with the appropriate size and power properties. The empirical results based on high-frequency data for a sample of Japanese stocks provide further evidence that the new testing procedure is capable of capturing various patterns of noise dependence. Thus, this statistical analysis of market microstructure noise sheds more light on the impact of market regularity and trading mechanism on asset pricing in financial markets.

## APPENDIX

### The proof of Lemma 1

Let the start and end times of intervals  $A_k$  and  $B_k$  be  $\underline{A}_k$ ,  $\overline{A}_k$  and  $\underline{B}_k$ ,  $\overline{B}_k$ ; that is,  $A_k = (\underline{A}_k, \overline{A}_k]$  and  $B_k = (\underline{B}_k, \overline{B}_k]$ . We consider the dependence between  $Z_{\ell,k}$  and  $Z_{\ell,k+h}$  for any  $h$  such that  $\underline{B}_{k+h} - \overline{A}_k \geq 0$ . It is obvious that  $Z_{\ell,k}$  has finite dependence from (1b) and (1d) in Assumption 1. Although there are some central limit theorems for finite dependence, as in Hoeffding and Robbins (1948) and Serfling (1968), the results given by Theorem 3.1 in Politis, Romano and Wolf (1997) are applied because the present study addresses more general cases of noise dependence including the mixing sequences. Assumption 2 implies that the conditional variance of a standardized sample mean of  $\{Z_{\ell,k'+1}, \dots, Z_{\ell,k'+n}\}$  for any  $k'$  approaches the limiting value  $\sigma_{\ell,f}^2$ . The condition for the strong mixing coefficient in Theorem 3.1 of Politis, Romano, and Wolf (1997) is satisfied from (1b) in Assumption 1. Therefore, it suffices to show the following condition (C1) for the application of their central limit theorem.

$$(C1) \ E_{\mathbb{I}} |Z_{\ell,k}|^{2\beta} < \infty, \quad \text{for some } \beta > 1.$$

Let  $\Delta\eta(A_k) := \eta(\overline{A}_k) - \eta(\underline{A}_k)$  and  $\Delta\delta(B_k) := \delta(\overline{B}_k) - \delta(\underline{B}_k)$  be the differences between the noises on each interval  $A_k$  and  $B_k$ .  $Z_{\ell,k}$  is decomposed as:

$$\begin{aligned} Z_{\ell,k} &= (P_1(\overline{A}_k) - P_1(\underline{A}_k))(P_2(\overline{B}_k) - P_2(\underline{B}_k)) = \int_{A_k} \sigma_1(u) dW_1(u) \int_{B_k} \sigma_2(u) dW_2(u) \\ &\quad + \int_{A_k} \sigma_1(u) dW_1(u) \Delta\delta(B_k) + \int_{B_k} \sigma_2(u) dW_2(u) \Delta\eta(A_k) + \Delta\eta(A_k) \Delta\delta(B_k). \end{aligned} \quad (28)$$

Given the assumption that  $\sigma_1(t)$  and  $\sigma_2(t)$  are bounded, it follows that  $\sigma_1(t), \sigma_2(t) < C$ , where  $C$  is a constant. On each interval  $A_k$  and  $B_k$ ,

$$\begin{aligned} E_{\mathbb{I}} \left| \int_{A_k} \sigma_1(u) dW_1(u) \right|^{2\beta} &< |C|^{2\beta} E_{\mathbb{I}} \left| \int_{A_k} dW_1(u) \right|^{2\beta} < \infty \quad \text{and} \\ E_{\mathbb{I}} \left| \int_{B_k} \sigma_2(u) dW_2(u) \right|^{2\beta} &< \infty. \end{aligned}$$

Because  $A_k$  and  $B_k$  are non-overlapping, the high-order absolute moment of the first term in (28),

$$E_{\mathbb{I}} \left| \int_{A_k} \sigma_1(u) dW_1(u) \int_{B_k} \sigma_2(u) dW_2(u) \right|^{2\beta} = E_{\mathbb{I}} \left| \int_{A_k} \sigma_1(u) dW_1(u) \right|^{2\beta} E_{\mathbb{I}} \left| \int_{B_k} \sigma_2(u) dW_2(u) \right|^{2\beta},$$

is bounded. From (1c) in Assumption 1 and Minkowski's inequality

$$E_{\mathbb{I}} |\Delta\delta(B_k)|^{2\beta} \leq (\|\delta(\overline{B}_k)\|_{2\beta} + \|\delta(\underline{B}_k)\|_{2\beta})^{2\beta} < \infty,$$

where  $\|\delta(\cdot)\|_{2\beta} = (E_{\Pi}|\delta(\cdot)|^{2\beta})^{\frac{1}{2\beta}}$ . From (1d) in Assumption 1, the high-order absolute moment of the second term in (28) has:

$$E_{\Pi} \left| \int_{A_k} \sigma_1(u) dW_1(u) \Delta\delta(B_k) \right|^{2\beta} = E_{\Pi} \left| \int_{A_k} \sigma_1(u) dW_1(u) \right|^{2\beta} E_{\Pi} |\Delta\delta(B_k)|^{2\beta} < \infty.$$

For the third term of (28),  $E_{\Pi} \left| \int_{B_k} \sigma_2(u) dW_2(u) \Delta\eta(A_k) \right|^{2\beta} < \infty$ . From (1c) in Assumption 1, the high-order absolute moment of the fourth term of (28) has:

$$\begin{aligned} & E_{\Pi} |\Delta\eta(A_k) \Delta\delta(B_k)|^{2\beta} \\ &= E_{\Pi} |\eta(\overline{A_k})\delta(\overline{B_k}) - \eta(\overline{A_k})\delta(\underline{B_k}) - \eta(\underline{A_k})\delta(\overline{B_k}) + \eta(\underline{A_k})\delta(\underline{B_k})|^{2\beta} \\ &\leq (\|\eta(\overline{A_k})\delta(\overline{B_k})\|_{2\beta} + \|\eta(\overline{A_k})\delta(\underline{B_k})\|_{2\beta} + \|\eta(\underline{A_k})\delta(\overline{B_k})\|_{2\beta} \\ &\quad + \|\eta(\underline{A_k})\delta(\underline{B_k})\|_{2\beta})^{2\beta} < \infty. \end{aligned}$$

Finally, we have:

$$\begin{aligned} E_{\Pi} |Z_{\ell,k}|^{2\beta} &\leq \left( \left\| \int_{A_k} \sigma_1(u) dW_1(u) \int_{B_k} \sigma_2(u) dW_2(u) \right\|_{2\beta} + \left\| \int_{A_k} \sigma_1(u) dW_1(u) \Delta\delta(B_k) \right\|_{2\beta} \right. \\ &\quad \left. + \left\| \int_{B_k} \sigma_2(u) dW_2(u) \Delta\eta(A_k) \right\|_{2\beta} + \|\Delta\eta(A_k) \Delta\delta(B_k)\|_{2\beta} \right)^{2\beta} < \infty. \quad (29) \end{aligned}$$

Condition (C1) holds. The asymptotic normality of  $f_{\ell, N_{\ell}}$  follows then from the central limit result in Politis, Romano, and Wolf (1997). ■

### The proof of Lemma 2

To demonstrate the consistency of the variance estimator in (7), it is possible to apply  $L_2$ -convergence of the subsampling estimator given by Lemma 4.6.1 in Politis, Romano, and Wolf (1999). Because the strong mixing condition holds from (1b) in Assumption 1, it suffices to show the following conditions (C2) and (C3) for the application of Lemma 4.6.1.

$$(C2) \quad K_{\ell}^{-1} \sum_{h=0}^{K_{\ell}-1} V_{\Pi}[M_{\ell}^{1/2} \bar{Z}_{\ell, M_{\ell}}^h] \rightarrow \sigma_{\ell, f}^2 \quad \text{s.t. } M_{\ell} \rightarrow \infty \text{ and } M_{\ell}/N_{\ell} \rightarrow 0 \text{ as } N_{\ell} \rightarrow \infty.$$

$$(C3) \quad (f_{\ell, N_{\ell}})^4 \text{ is uniformly integrable.}$$

Denote  $V_{\Pi}[M_{\ell}^{1/2} \bar{Z}_{\ell, M_{\ell}}^h]$  as  $\sigma_{\ell, h}^2$ . Then we have:

$$\frac{1}{K_{\ell}} \sum_{h=0}^{K_{\ell}-1} \sigma_{\ell, h}^2 - \sigma_{\ell, f}^2 \leq \frac{1}{K_{\ell}} \sum_{h=0}^{K_{\ell}-1} |\sigma_{\ell, h}^2 - \sigma_{\ell, f}^2| = \sup_{0 \leq h \leq K_{\ell}-1} |\sigma_{\ell, h}^2 - \sigma_{\ell, f}^2| \rightarrow 0$$

s.t.  $M_{\ell} \rightarrow \infty$  as  $N_{\ell} \rightarrow \infty$  from Assumption 2. Thus, (C2) holds.

For (C3), it is possible to show that  $E_{\Pi} |Z_{\ell, k}|^{4\beta} < \infty$  for some  $\beta > 1$  from (1c) in Assumption 1 by the similar argument as the proof of (C1). Let the centered  $Z_{\ell, k}$  be  $Z_{\ell, k}^*$ . It is obvious that  $E_{\Pi} |\sum_{k=1}^{N_{\ell}} Z_{\ell, k}^*|^{4\beta} = O(N_{\ell}^{2\beta})$  because the sequence  $\{Z_{\ell, k}^*\}$  is  $m$ -dependent with  $E_{\Pi} |Z_{\ell, k}^*|^{4\beta} < \infty$ . Therefore, the order of

$E_{\Pi}|f_{\ell, N_{\ell}}|^{4\beta} = E_{\Pi}|N_{\ell}^{-1/2} \sum_{k=1}^{N_{\ell}} Z_{\ell, k}^*|^{4\beta}$  becomes  $O(1)$ . Then (C3) holds because  $E_{\Pi}|f_{\ell, N_{\ell}}|^{4\beta} < \infty$  implies that  $(f_{\ell, N_{\ell}})^4$  is uniformly integrable. Finally, these results yield  $\hat{\sigma}_{\ell, f}^2 \xrightarrow{a} \sigma_{\ell, f}^2$  as  $N_{\ell} \rightarrow \infty$ . ■

### The proof of Theorem 2

The conditional expectation of  $\hat{\gamma}(\ell)$  is:

$$E_{\Pi}[\hat{\gamma}(\ell)] = -\frac{1}{N_{\ell}^{(\pm)}} \sum_{k=1}^{N_{\ell}^{(\pm)}} E_{\Pi}[Z_{\ell, k}^{(\pm)}] = \gamma(\ell). \quad (30)$$

For the conditional variance of  $\hat{\gamma}(\ell)$ , it can be shown that:

$$\begin{aligned} V_{\Pi}[\hat{\gamma}(\ell)] &= V_{\Pi} \left[ -\frac{1}{N_{\ell}^{(\pm)}} \sum_{k=1}^{N_{\ell}^{(\pm)}} Z_{\ell, k}^{(\pm)} \right] \\ &= \frac{1}{(N_{\ell}^{(\pm)})^2} \sum_{k=1}^{N_{\ell}^{(\pm)}} V_{\Pi}[Z_{\ell, k}^{(\pm)}] + \frac{2}{(N_{\ell}^{(\pm)})^2} \sum_{k=1}^{N_{\ell}^{(\pm)}} \sum_{j=1}^{N_{\ell}^{(\pm)}-k} \text{Cov}_{\Pi}[Z_{\ell, k}^{(\pm)}, Z_{\ell, k+j}^{(\pm)}] \\ &\leq \frac{1}{(N_{\ell}^{(\pm)})^2} \sum_{k=1}^{N_{\ell}^{(\pm)}} \max_k \{V_{\Pi}[Z_{\ell, k}^{(\pm)}]\} + \frac{2}{(N_{\ell}^{(\pm)})^2} \sum_{k=1}^{N_{\ell}^{(\pm)}} \sum_{j=1}^{\tilde{m}_k} \max_k \{|\text{Cov}_{\Pi}[Z_{\ell, k}^{(\pm)}, Z_{\ell, k+j}^{(\pm)}]|\} \\ &= O\left(\frac{1}{N_{\ell}^{(\pm)}}\right), \end{aligned} \quad (31)$$

where  $\tilde{m}_k$  is defined as  $\max_j \{\text{Cov}_{\Pi}[Z_{\ell, k}^{(\pm)}, Z_{\ell, k+j}^{(\pm)}] \neq 0, 0 < j \leq N_{\ell}^{(\pm)} - k\}$ . Because of the finite dependence of  $\{Z_{\ell, k+j}^{(\pm)}\}_{k=1}^{N_{\ell}^{(\pm)}}$ ,  $\tilde{m}_k$  is finite. This implies  $V_{\Pi}[\hat{\gamma}(\ell)] \rightarrow 0$  as  $N_{\ell}^{(\pm)}$  goes to infinity and the consistency of  $\hat{\gamma}(\ell)$  holds. Let the asymptotic variance of  $\hat{\gamma}(\ell)$  be  $\omega_{\ell}^2 = \lim_{N_{\ell}^{(\pm)} \rightarrow \infty} E_{\Pi}[N_{\ell}^{(\pm)}(\hat{\gamma}(\ell) - \gamma(\ell))^2]$ . The asymptotic normality of  $\hat{\gamma}(\ell)$  can be demonstrated by a similar argument to the proof of Lemma 1. The difference between  $\{Z_{\ell, k}^{(\pm)}\}_{k=1}^{N_{\ell}^{(\pm)}}$  and  $\{Z_{\ell, k}\}_{k=1}^{N_{\ell}}$  represents the degree of dependence. The sequence of  $\{Z_{\ell, k}^{(\pm)}\}_{k=1}^{N_{\ell}^{(\pm)}}$  has higher dependence than  $\{Z_{\ell, k}\}_{k=1}^{N_{\ell}}$  because  $Z_{\ell, k}^{(\pm)}$  represents the product of returns on nonoverlapping intervals, with each interval being inclusive and extending beyond  $A_k$  and  $B_k$ . However,  $Z_{\ell, k}^{(\pm)}$  has finite dependence from (1b) and (1d) in Assumption 1. ■

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